人工智能学院课程——现代控制理论

卡尔曼滤波算法介绍

薛文超

中国科学院数学与系统科学研究院

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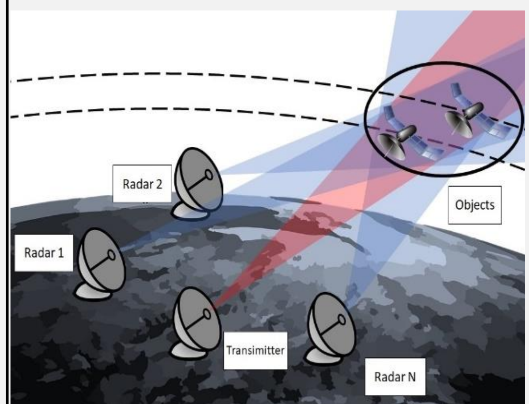
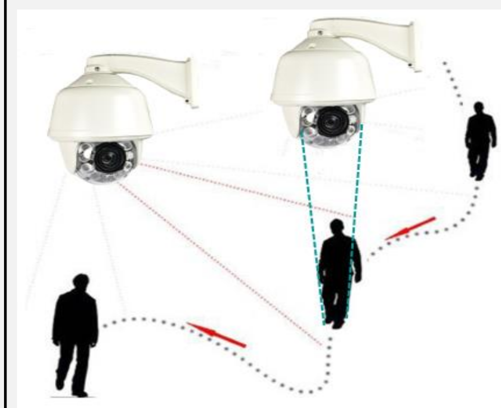
wenchaoxue@amss.ac.cn

概要

- 研究背景和问题描述
- 卡尔曼滤波算法及其主要特性
- 卡尔曼滤波算法的一致性
- 扩张状态卡尔曼滤波算法

研究背景

滤波：最优的状态估计

	Volatility Estimation	Orbit Determination	Camera Tracking
实际系统			
Observation data	Disturbed stock price	Radar measurements	Measurements of sensors in network
System state	Volatility	Position and velocity	Position and velocity

目标跟踪算法：来源于雷达数据处理

□ 雷达从二次大战开始被广泛使用

- 1936年1月英国沃森·瓦特在索夫克海岸架起了英国第1个雷达站
- 1937年美国第一个军舰雷达XAF试验成功
- 1941年苏联最早在飞机上装备预警雷达

□ 雷达发挥了重要作用：不列颠之战、珍珠港、菲律宾、诺曼底

- 沿海雷达站的连锁装置为成功拦截敌机作出了巨大贡献
- 雷达在盟军军舰上的率先使用，军舰不再轻易成为飞机的猎物

本土链雷达系统(Chain home Radar System), 1940s, 英国

目标跟踪算法：来源于雷达数据处理

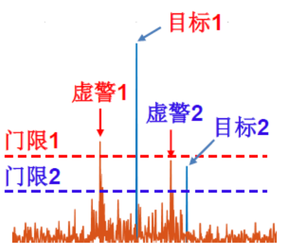
雷达工作原理：发射 → 反射 → 接收 → 处理

- 发射：雷达发射电磁脉冲
- 反射：目标反射电磁脉冲
- 接收：雷达接收目标反射脉冲
- 处理：根据反射脉冲，检测目标有无、测量目标信息

☐ 目标检测

理论基础之一： 1948, Shannon, 通信中的数学原理

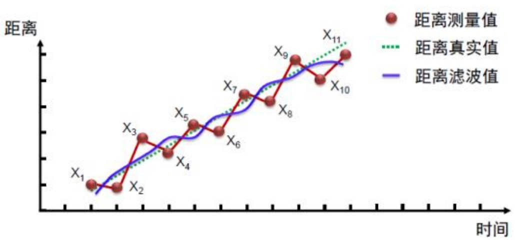
检测的任务：降低虚警率和漏警率



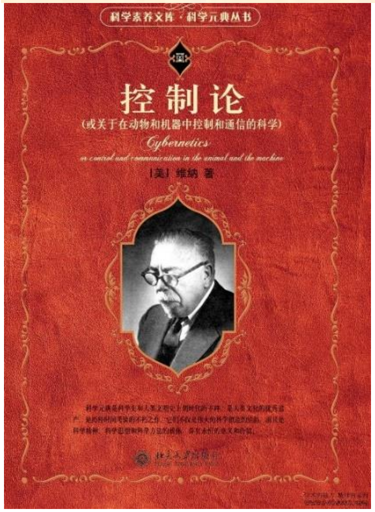
☐ 目标跟踪

理论基础之一： 1940's, Weiner, 滤波与控制

跟踪的任务：平滑与预报



控制论：科学的一门重要分支

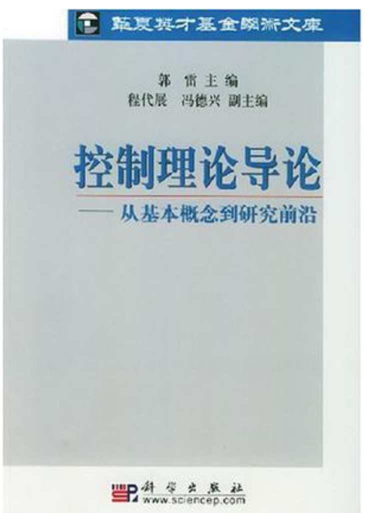


- **诞生标志：**

  关于在动物和机器中控制和通讯的科学（维纳，1948）

  - 时间序列、信息与通信
  - 维纳的预测与滤波理论
  - 反馈控制原理与稳定性
  - 计算机与神经系统
  - 控制论与心理病理学
  - 信息、语言与社会
  - …

控制理论的范畴



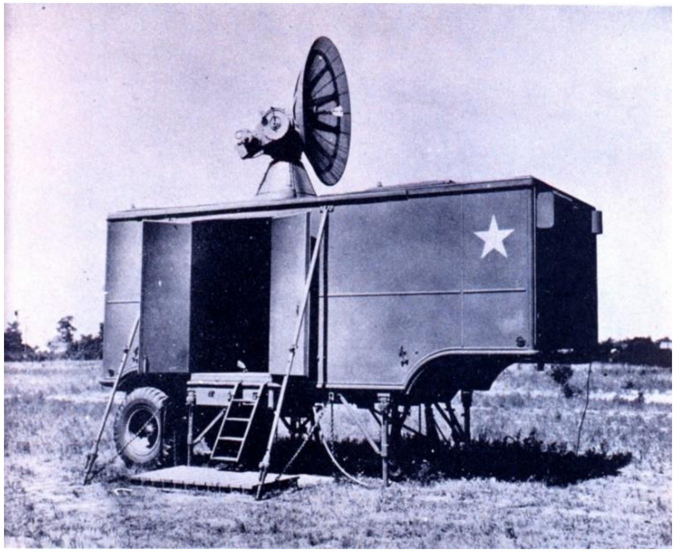
- **经典控制理论**
  - 传递函数与频域分析方法…
- **现代控制理论**
  - 微分方程
  - 概率论
  - 微分几何
  - 泛函分析
  - 稳定性理论
  - 最优控制理论
  - 最优估计理论
  - …

1970-1980年代控制室编写的
现代控制系统理论小丛书系

雷达边扫描边跟踪系统(track-while-scan systems)

□ 1940s：迫切需求——处理原始雷达信号以识别目标和更新轨迹

早期人工处理：首次探测到火控系统精确跟踪需**几十秒至几分钟，效率较低！**



发展自动的边扫描边跟踪系统

- 目标检测/识别
- 目标轨迹关联
- 目标轨迹初始化与生成
- 生成轨迹跟踪门限
- 平滑（滤波）和定位
- 计算和显示目标未来位置

Filter

SCR-584：麻省理工辐射实验室研发的最著名雷达之一（火控系统）
1944 年在意大利的安齐奥滩头阵地首次用于战争

□ 1960s：相控阵雷达和计算机存储设备使雷达自动检测和跟踪成为可能

自动处理：首次探测到火控系统精确跟踪需**几秒钟，效率大幅提高！**

从Wiener Filter到$\alpha-\beta-\gamma$ Filer

Wiener Filter问题：基于当时时刻及之前量测数据得到线性最优估计值

第n时刻信号真值为x_n, 量测数据为$\overline{x}_n$, 量测噪声序列为i.i.d.

通常估计值表达为量测数据的线性组合$\sum_{k=0}^{n} g_k\overline{x}_{n-k}$，关心的优化指标：

$$\min_{g_k} E\left\{\left(x_n-\sum_{k=0}^{n} g_k\overline{x}_{n-k}\right)^2\right\}$$

最小化估计误差均方差

计算过程能否实现实时递推?

□ 1962：二阶系统（匀速运动）

通过对指标函数的适当调整

位置估计值： $\overline{x}_n = x_{pn}+\alpha(x_n-x_{pn})$

速度估计值： $\dot{\overline{x}}_n = \dot{\overline{x}}_{n-1}+\frac{\beta}{T}(x_n-x_{pn})$

位置预报值： $x_{p\,n+1} = \overline{x}_n+T\dot{\overline{x}}_n$

最优设计： $\beta=\alpha^2/(2-\alpha)$

□ 1963：三阶系统（匀加速运动）

通过对指标函数的适当调整

位置估计值： $\overline{x}_n = x_{pn}+\alpha(x_n-x_{pn})$

速度估计值： $\dot{\overline{x}}_n = \dot{\overline{x}}_{n-1}+T\ddot{\overline{x}}_{n-1}+\frac{\beta}{T}(x_n-x_{pn})$

加速度估计值： $\ddot{\overline{x}}_n = \ddot{\overline{x}}_{n-1}+\frac{\gamma}{T^2}(x_n-x_{pn})$

位置预报值： $x_{p\,n+1} = \overline{x}_n+T\dot{\overline{x}}_n+\frac{1}{2}T^2\ddot{\overline{x}}_n$

最优设计： $2\beta-\alpha(\alpha+\beta+\gamma/2)=0$

卡尔曼滤波：超时代的统一算法

卡尔曼（Rudolf Emil Kalman，1930-2016），出生于匈牙利布达佩斯，数学家，法国科学院外籍院士，美国国家工程院院士，美国艺术与科学院院士等。

卡尔曼最重要的发明是卡尔曼滤波算法，该算法成就了过去50年间的许多基本技术，如把阿波罗号宇航员送上月球的航天计算机系统、把人类送去探索深海和星球的机器人载体，以及几乎所有需要从噪声数据去估算实际状态的项目。有人甚至把包括环绕地球的卫星系统、卫星地面站及各类计算机系统在内的整个GPS系统合称为一个巨大无比的卡尔曼滤波器。

David Mindell, FrankMoss, How an Inventor You've Probably Never Heard of Shaped the Modern World, MIT Review, 2016.

准备知识：随机变量

- **随机变量：随机试验各种结果的实值单值函数**

  X：将随机试验各种结果映射到实数

  (扔硬币出现正面记为1，扔硬币出现反面记为0)

- **随机变量X最基本的特性：概率分布函数**

  (probability distribution function, PDF) $F_X(x) = P(X \leq x)$

- **PDF满足的特性：**

$$
\begin{aligned}
F_X(x) &\in [0,1] \\
F_X(-\infty) &= 0 \\
F_X(\infty) &= 1 \\
F_X(a) &\leq F_X(b) \quad \text{if } a \leq b \\
P(a < X \leq b) &= F_X(b) - F_X(a)
\end{aligned}
$$

准备知识：概率分布函数(PDF)和概率密度函数(pdf)

- **概率密度函数(Probability density function, pdf)：PDF的导数**

$$f_X(x) = \frac{dF_X(x)}{dx}$$

- **pdf满足的特性：**

$$\begin{aligned}
F_X(x) &= \int_{-\infty}^{x} f_X(z)\,dz \\
f_X(x) &\geq 0 \\
\int_{-\infty}^{\infty} f_X(x)\,dx &= 1 \\
P(a < x \leq b) &= \int_{a}^{b} f_X(x)\,dx
\end{aligned}$$

准备知识：条件概率分布函数和条件概率密度函数

- 给定事件A条件下的概率分布

$$
\begin{aligned}
F_X(x|A) &= P(X \le x|A) \\
&= \frac{P(X \le x, A)}{P(A)}
\end{aligned}
$$

- 给定事件A条件下的概率密度

$$
f_X(x|A) = \frac{dF_X(x|A)}{dx}
$$

准备知识：期望和方差

- g(X)为随机变量的一个函数（将随机变量映射为随机变量），则g(X)的期望为 $E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\,dx$

- X的期望为： $\bar{x} \triangleq E(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx$

- X的方差：
$$\begin{aligned}\sigma_X^2 &= E[(X-\bar{x})^2] \\ &= \int_{-\infty}^{\infty} (x-\bar{x})^2 f_X(x)\,dx\end{aligned}$$

- 常用标记： $X \sim (\bar{x}, \sigma^2)$

- **条件期望：** $E(X|A) = \int_{-\infty}^{\infty} x f_X(x|A)\,dx$

准备知识：高维随机变量

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ F_Y(y) &= P(Y \leq y) \end{aligned}$$

- **联合分布函数：** $F_{XY}(x,y) = P(X \leq x, Y \leq y)$

(扔两个硬币，出现正面+1，出现反面+0)

$$\begin{aligned} F(x,y) &\in [0,1] \\ F(x,-\infty) = F(-\infty,y) &= 0 \\ F(\infty,\infty) &= 1 \\ F(a,c) &\leq F(b,d) \quad \text{if } a \leq b \text{ and } c \leq d \\ P(a < x \leq b, c < y \leq d) &= F(b,d) + F(a,c) - F(a,d) - F(b,c) \\ F(x,\infty) &= F(x) \\ F(\infty,y) &= F(y) \end{aligned}$$

准备知识：高维密度函数

➢ **联合密度函数：** $f_{XY}(x,y) = \dfrac{\partial^2 F_{XY}(x,y)}{\partial x \partial y}$

$$
\begin{aligned}
F(x,y) &= \int_{-\infty}^{x} \int_{-\infty}^{y} f(z_1, z_1)\, dz_1\, dz_2 \\
f(x,y) &\geq 0 \\
\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\, dx\, dy &= 1 \\
P(a < x \leq b, c < y \leq d) &= \int_{c}^{d} \int_{a}^{b} f(x,y)\, dx\, dy \\
f(x) &= \int_{-\infty}^{\infty} f(x,y)\, dy \\
f(y) &= \int_{-\infty}^{\infty} f(x,y)\, dx
\end{aligned}
$$

准备知识：高维密度函数

➢ **边际密度和分布函数：**

$$f_Y(y)=\int_{-\infty}^{\infty} f(x,y)dx \qquad F_Y(y)=\int_{-\infty}^{y}\int_{-\infty}^{\infty} f(x,t)dxdt$$

$$f_X(x)=\int_{-\infty}^{\infty} f(x,y)dy \qquad F_X(y)=\int_{-\infty}^{y}\int_{-\infty}^{\infty} f(t,y)dydt$$

➢ **条件概率密度和条件概率分布**

$$f_{X|Y}(x\,|\,\mathrm{y})=\frac{f(x,\mathrm{y})}{f_Y(y)}$$

$$F_{X|Y}(X\,|\,Y)=\int_{-\infty}^{x} f_{X|Y}(t\,|\,\mathrm{y})\mathrm{dt}=\int_{-\infty}^{x}\frac{f(t,\mathrm{y})}{f_Y(y)}\mathrm{dt}$$

准备知识：独立性

- **随机变量X和Y独立：** $P(X \le x, Y \le y) = P(X \le x)P(Y \le y) \quad \text{for all } x, y$
- **随机变量X和Y独立：** $$\begin{aligned} F_{XY}(x,y) &= F_X(x)F_Y(y) \\ f_{XY}(x,y) &= f_X(x)f_Y(y) \end{aligned}$$
- **独立时的条件概率密度** $f_{X|Y}(x \mid y) = \dfrac{f(x,y)}{f_Y(y)} = f_X(x)$

准备知识：高维随机变量的相关函数

➢ **互相关函数矩阵：**

$$
\begin{aligned}
R_{XY} &= E(XY^T) \\
&= \begin{bmatrix} E(X_1Y_1) & \cdots & E(X_1Y_m) \\ \vdots & & \vdots \\ E(X_nY_1) & \cdots & E(X_nY_m) \end{bmatrix}
\end{aligned}
$$

➢ **互协方差矩阵：**

$$
\begin{aligned}
C_{XY} &= E[(X-\bar{X})(Y-\bar{Y})^T] \\
&= E(XY^T) - \bar{X}\bar{Y}^T
\end{aligned}
$$

准备知识：高维随机变量的协方差阵

➢ **自相关函数矩阵**

$$
\begin{aligned}
R_X &= E[XX^T] \\
&= \begin{bmatrix} E[X_1^2] & \cdots & E[X_1X_n] \\ \vdots & & \vdots \\ E[X_nX_1] & \cdots & E[X_n^2] \end{bmatrix}
\end{aligned}
$$

➢ **自协方差矩阵**

$$
\begin{aligned}
C_X &= E[(X-\bar{X})(X-\bar{X})^T] \\
&= \begin{bmatrix} E[(X_1-\bar{X}_1)^2] & \cdots & E[(X_1-\bar{X}_1)(X_n-\bar{X}_n)] \\ \vdots & & \vdots \\ E[(X_n-\bar{X}_n)(X_1-\bar{X}_1)] & \cdots & E[(X_n-\bar{X}_n)^2] \end{bmatrix} \\
&= \begin{bmatrix} \sigma_1^2 & \cdots & \sigma_{1n} \\ \vdots & & \vdots \\ \sigma_{n1} & \cdots & \sigma_n^2 \end{bmatrix}
\end{aligned}
$$

自协方差矩阵为半正定矩阵

$$
\begin{aligned}
z^T C_X z &= z^T E[(X-\bar{X})(X-\bar{X})^T] z \\
&= E[z^T(X-\bar{X})(X-\bar{X})^T z] \\
&= E[(z^T(X-\bar{X}))^2] \\
&\geq 0
\end{aligned}
$$

Note that $\sigma_{ij} = \sigma_{ji}$ so $C_X = C_X^T$.

准备知识：随机过程

随机过程：一列随机变量X(t)，t为时间

均为时变的函数：

$$F_X(x,t) = P(X(t) \le x)$$

$$F_X(x,t) = P\left[X_1(t) \le x_1 \text{ and } \cdots X_n(t) \le x_n(t)\right]$$

$$f_X(x,t) = \frac{d^n F_X(x,t)}{dx_1 \cdots dx_n}$$

$$\bar{x}(t) = \int_{-\infty}^{\infty} x f(x,t)\, dx$$

$$\begin{aligned} C_X(t) &= E\left\{[X(t) - \bar{x}(t)]\,[X(t) - \bar{x}(t)]^T\right\} \\ &= \int_{-\infty}^{\infty} [x - \bar{x}(t)]\,[x - \bar{x}(t)]^T f(x,t)\, dx \end{aligned}$$

$$R_X(t_1,t_2) = E\left[X(t_1)X^T(t_2)\right]$$

$$C_X(t_1,t_2) = E\left\{\left[X(t_1) - \bar{X}(t_1)\right]\left[X(t_2) - \bar{X}(t_2)\right]^T\right\}$$

离散时变线性系统的滤波问题

$$\begin{cases} X_{k+1} = A_k X_k + B_k u_k + w_k \\ Y_{k+1} = C_{k+1} X_{k+1} + n_{k+1} \end{cases}$$

$$k = 0,1,2,\ldots$$

$\{n_k\}$和$\{w_k\}$为随机过程

$$E(n_k) = 0, \forall k > 0$$

$$E(w_k) = 0, \forall k > 0$$

$$C_n(k, j) = \begin{cases} 0, \forall k \neq j \\ R_k, \forall k = j \end{cases}$$

$$C_w(k, j) = \begin{cases} 0, \forall k \neq j \\ Q_k, \forall k = j \end{cases}$$

离散时变线性系统的滤波问题

$$\begin{cases} X_{k+1} = A_k X_k + B_k u_k + w_k \\ Y_{k+1} = C_{k+1} X_{k+1} + n_{k+1} \end{cases}$$

$$k = 0, 1, 2, \ldots$$

$\{n_k\}$和$\{w_k\}$为随机过程

$$E(n_k) = 0, \forall k > 0$$

$$E(w_k) = 0, \forall k > 0$$

$$C_n(k, j) = \begin{cases} 0, \forall k \neq j \\ R_k, \forall k = j \end{cases}$$

$$C_w(k, j) = \begin{cases} 0, \forall k \neq j \\ Q_k, \forall k = j \end{cases}$$

目标：利用量测 $\{Y_i, 1 \leq i \leq k\}$获得 X_k 的线性最小方差估计

$$\hat{X}_k = \underset{Z_k \in \{\text{linear function of } Y_i, 1 \leq i \leq k\}}{\arg\min} E\left\{(Z_k - X_k)(Z_k - X_k)^T\right\}$$

概要

- 研究背景和问题描述
- 卡尔曼滤波算法及其主要特性
- 卡尔曼滤波算法的一致性
- 扩张状态卡尔曼滤波算法

卡尔曼滤波算法及参数设计

离散线性系统：

$$\begin{cases} X_{k+1} = A_k X_k + B_k u_k + w_k \\ Y_{k+1} = C_{k+1} X_{k+1} + n_{k+1} \end{cases}$$

$$k = 0,1,2,\ldots$$

1930-2016

卡尔曼滤波(1960s, Kalman):

初始化过程：

$$\begin{cases} P_0 = E\left(X_0 - EX_0\right)\left(X_0 - EX_0\right)^T \\ \hat{X}_0 = EX_0 \end{cases}$$

参数选择：

$$\begin{cases} Q_k = E\left(w_k w_k^T\right) \\ R_{k+1} = E\left(n_{k+1} n_{k+1}^T\right) > 0 \end{cases}$$

$\bar{X}_{k+1} = A_k \hat{X}_k + B_k u_k$： 预测过程

$\bar{P}_{k+1} = A_k P_k A_k^T + Q_k$： 预测误差的协方差阵

$\hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}(Y_{k+1} - C_{k+1}\bar{X}_{k+1})$：更新过程

$K_{k+1} = (\bar{P}_{k+1} C_{k+1}^T)(C_{k+1}\bar{P}_{k+1}C_{k+1}^T + R_{k+1})^{-1}$： 更新增益

$P_{k+1} = \bar{P}_{k+1} - K_{k+1} C_{k+1} \bar{P}_{k+1}$： 估计/滤波误差的协方差阵

考虑线性定常系统：

$$\begin{cases} X_{k+1} = AX_k + Bu_k + w_k \\ Y_{k+1} = CX_{k+1} + n_{k+1} \end{cases}$$

$$k = 0,1,2,\ldots$$

$$\begin{cases} \hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}(Y_{k+1} - C\bar{X}_{k+1}) \\ \bar{X}_{k+1} = A\hat{X}_k + Bu_k \\ K_{k+1} = (\bar{P}_{k+1}C^T)(C\bar{P}_{k+1}C^T + R)^{-1} \\ P_{k+1} = \bar{P}_{k+1} - K_{k+1}C\bar{P}_{k+1} \\ \bar{P}_{k+1} = AP_kA^T + Q \end{cases}$$

定理：卡尔曼滤波满足如下特性：

1. 线性最小方差估计

2. $P_k = \mathbb{E}\left\{(X_k - \hat{X}_k)(X_k - \hat{X}_k)^T\right\}$

3.当(A,C)可观时，滤波误差均方有界

定理的证明需要下面3个引理

引理1. 设x和y分别是具有前二阶矩的随机向量，则利用y对x的线性无偏最小方差估计为

$$\hat{x}_y = \underbrace{\mathbb{E}\{x\}}_{\text{x变量本身的期望}} + \underbrace{R_{xy}R_y^{-1}}_{\text{增益}}\underbrace{(y - \mathbb{E}\{y\})}_{\text{y带来的新信息}},$$

其中

$$R_{xy} \triangleq \mathbb{E}\left\{(x - \mathbb{E}\{x\})(y - \mathbb{E}\{y\})^T\right\},$$

$$R_y \triangleq \mathbb{E}\{(y - \mathbb{E}\{y\})(y - \mathbb{E}\{y\})^T\}.$$

引理1的证明：由估计的线性无偏性可知$\hat{x}_y$具有如下形式：

$$\hat{x}_y = \mathbb{E}\{x\} + K(y - \mathbb{E}\{y\}).$$

由此易得

$$\mathbb{E}\left\{\left(\hat{x}_y - x\right)\left(\hat{x}_y - x\right)^T\right\}$$

$$= \mathbb{E}\{(x - \mathbb{E}\{x\} - K(y - \mathbb{E}\{y\}))(x - \mathbb{E}\{x\} - K(y - \mathbb{E}\{y\}))^T\}$$

$$= R_x - KR_{yx} - R_{xy}K^T + KR_yK^T$$

$$= \left(K - R_{xy}R_y^{-1}\right)R_y\left(K - R_{xy}R_y^{-1}\right)^T + R_x - R_{xy}R_y^{-1}R_{yx}$$

其中最后一个等式的第1项为非负定项，故欲令估计误差的协方差阵最小，当且仅当$K = R_{xy}R_y^{-1}$.

引理2. 设u为已知向量，x和y是具有前二阶矩的随机向量，记y对x的线性无偏最小方差估计为$\hat{x}_y$, 则利用y对$z \triangleq Ax + Bu + Cy$的线性无偏最小方差估计为$\hat{z}_y = A\hat{x}_y + Bu + Cy$.

引理2的证明：

无偏性需$\hat{z}_y$ 的形式为: $\mathrm{A(E\{}x\mathrm{\} + K(y - E\{y\})} + Bu + \mathrm{Cy}$

最小方差需要：$\hat{z}_y = A\hat{x}_y + Bu + Cy$

引理3. 设x, $\mathscr{y}$和y是具有前二阶矩的随机向量，记 $\mathrm{w} \triangleq [\mathscr{y}^T \quad y^T]^T$, 若利用$\mathscr{y}$对x和y的线性无偏最小方差估计分别为$\hat{x}_{\mathscr{y}}$和$\hat{y}_{\mathscr{y}}$，则利用w对x的线性无偏最小方差估计满足

$\mathrm{w} \triangleq [\mathscr{y}^T \quad \mathrm{y}^T]^T$：所有数据

y：当前数据

$\mathscr{y}$：过去数据

$$\hat{x}_{\mathrm{w}} = \hat{x}_{\mathscr{y}} + \mathrm{K}(\mathrm{y} - \hat{y}_{\mathscr{y}}),$$

利用过去数据得到的估计

新息增益

当前数据带来的新息

其中

$$\mathrm{K} = R_{\mathrm{xy}|\mathscr{y}} R_{\mathrm{y}|\mathscr{y}}^{-1},$$

$$R_{\mathrm{xy}|\mathscr{y}} \triangleq \mathbb{E}\{(\mathrm{x} - \hat{x}_{\mathscr{y}})(\mathrm{y} - \hat{y}_{\mathscr{y}})^T\},$$

$$R_{\mathrm{y}|\mathscr{y}} \triangleq \mathbb{E}\{(\mathrm{y} - \hat{y}_{\mathscr{y}})(\mathrm{y} - \hat{y}_{\mathscr{y}})^T\}.$$

引理3的证明：

引入新变量：$e \triangleq x - \hat{x}_{\mathcal{Y}} - K(y - \hat{y}_{\mathcal{Y}})$, 其中$\mathrm{K} = R_{\mathrm{xy}|\mathcal{Y}} R_{\mathrm{y}|\mathcal{Y}}^{-1}$，则容易验证$\mathrm{E}\{e\} = 0$. 接下来分析e的进一步性质：

由引理1可得：

$$\hat{x}_{\mathcal{Y}} = \mathrm{E}\{\mathrm{x}\} + R_{x\mathcal{Y}} R_{\mathcal{Y}\mathcal{Y}}^{-1}(\mathcal{Y} - E\{\mathcal{Y}\})$$

$$\hat{y}_{\mathcal{Y}} = \mathrm{E}\{y\} + R_{\mathrm{y}\mathcal{Y}} R_{\mathcal{Y}\mathcal{Y}}^{-1}(\mathcal{Y} - E\{\mathcal{Y}\})$$

将上面两个式子带入到e的表达式中：

$$e = x - \mathrm{E}\{\mathrm{x}\} - R_{x\mathcal{Y}} R_{\mathcal{Y}\mathcal{Y}}^{-1}(\mathcal{Y} - E\{\mathcal{Y}\}) - R_{xy|\mathcal{Y}} R_{\mathrm{y}|\mathcal{Y}}^{-1}(y - \mathrm{E}\{y\} - R_{\mathrm{y}\mathcal{Y}} R_{\mathcal{Y}\mathcal{Y}}^{-1}(\mathcal{Y} - E\{\mathcal{Y}\}))$$

进一步再分析e与其它变量的相关函数：首先，容易验证

$$\mathrm{E}\{e(\mathcal{Y} - E\{\mathcal{Y}\})^T\} = 0$$

此外，还根据$R_{xy|\mathcal{Y}} = R_{xy|\mathcal{Y}} R_{\mathrm{y}|\mathcal{Y}}^{-1} R_{\mathrm{y}|\mathcal{Y}}$ 和 $e \triangleq x - \hat{x}_{\mathcal{Y}} - R_{xy|\mathcal{Y}} R_{\mathrm{y}|\mathcal{Y}}^{-1}(y - \hat{y}_{\mathcal{Y}})$ 得到

$$\mathrm{E}\{e(\mathrm{y} - \hat{y}_{\mathcal{Y}})\} = 0.$$

引理3的证明：

进一步由 $\mathrm{E}\{e(\mathcal{Y}-E\{\mathcal{Y}\})\}=0$ 和 $\mathrm{E}\{e(\mathrm{y}-\hat{y}_y)\}=0$ 以及

$$y-E\{y\}=\mathrm{y}-\hat{y}_y-R_{\mathrm{y}\mathcal{Y}}R_{\mathcal{Y}\mathcal{Y}}^{-1}(\mathcal{Y}-E\{\mathcal{Y}\}))$$

可得

$$\mathrm{E}\{e(\mathrm{y}-\mathrm{E}\{y\})\}=0,$$

从而有$R_{ew}\triangleq\mathbb{E}\{e(w-\mathbb{E}\{w\})^T\}=\mathbf{0}$. 设$\hat{e}_w$为w对e的线性最小方差估计，则根据引理1知$\hat{e}_w=0.$

根据e的定义有：

$$x=\mathrm{e}+\hat{x}_y+K(y-\hat{y}_y),$$

设$\hat{x}_w$为w对x的线性最小方差估计，则由引理2可得：

$$\hat{x}_w=\hat{e}_w+\hat{x}_y+K(y-\hat{y}_y)=\hat{x}_y+K(y-\hat{y}_y).$$

证毕。

定理的证明

记$\mathcal{Y}_k \triangleq \begin{bmatrix} Y_0^T & Y_1^T & \cdots & Y_k^T \end{bmatrix}^T$对$X_k, X_{k+1}$和$Y_{k+1}$的线性无偏最小方差估计分别为$\hat{X}_k, \bar{X}_{k+1}$和$\bar{Y}_{k+1}$, 则根据引理2可得$\bar{X}_{k+1} = A\hat{X}_k + Bu_k$, $\bar{Y}_{k+1} = C\bar{X}_{k+1}$. 同时记$\hat{X}_k$和$\bar{X}_{k+1}$的估计误差协方差阵为$P_k$和$\bar{P}_{k+1}$, 则$\bar{P}_{k+1} = AP_kA^T + Q$. 进一步地，将**引理3中的$x, \mathcal{y}$和$y$分别替换成$X_{k+1}$, $\mathcal{Y}_k$和Y_{k+1}**, 可得

$$\hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}(Y_{k+1} - C\bar{X}_{k+1})$$

$$K_{k+1} = \mathbb{E}\{(X_{k+1} - \bar{X}_{k+1})(Y_{k+1} - \bar{Y}_{k+1})^T\}\left(\mathbb{E}\{(Y_{k+1} - \bar{Y}_{k+1})(Y_{k+1} - \bar{Y}_{k+1})^T\}\right)^{-1}$$

$$= \bar{P}_{k+1} C^T\left(C\bar{P}_{k+1}C^T + R\right)^{-1},$$

$$P_{k+1} = (\mathbf{I} - K_{k+1}C)\bar{P}_{k+1}(\mathbf{I} - K_{k+1}C)^T + K_{k+1}RK_{k+1}$$

$$= \bar{P}_{k+1} - K_{k+1}C\bar{P}_{k+1}.$$

定理的证明（续）

$$\begin{cases} X_{k+1} = AX_k + Bu_k + w_k \\ Y_{k+1} = CX_{k+1} + n_{k+1} \end{cases}$$

$k = 0,1,2,\ldots$

记$Y_{k(n)} \triangleq \begin{bmatrix} Y_{k-n+1}^T & Y_{k-n}^T & \cdots & Y_k^T \end{bmatrix}^T$, $M \triangleq$ $\begin{bmatrix} C^T & (CA)^T & \cdots & (CA^{n-1})^T \end{bmatrix}^T$, 由于$(A, C)$能观，亦即$M$是列满秩的，从而**可构造如下估计**（先假设$u_k$ =0,≠0时的证明类似）

$$\tilde{X}_k = \begin{cases} A^k \mathbb{E}\{x_0\}, & k < n \\ A^{n-1}\left(M^T M\right)^{-1} M^T Y_{k(n)}, & k \geq n \end{cases}$$

易得**$\tilde{X}_k$的估计误差协方差阵$\tilde{P}_k$是有界的**。

由于$\tilde{P}_k$是最小方差的线性估计，从而$P_k \leq \tilde{P}_k$也是有界的。

概要

- 研究背景和问题描述
- 卡尔曼滤波算法及其主要特性
- **卡尔曼滤波算法的一致性**
- 扩张状态卡尔曼滤波算法

线性系统的卡尔曼滤波

$$\begin{cases} X_{k+1} = AX_k + BF(X_k, k) + w_k \\ Y_{k+1} = CX_{k+1} + n_{k+1} \end{cases}$$

$$k = 0,1,2,\ldots$$

X_k : state, Y_k : measured output

(A, B, C): known matrices

$F(X_k, k)$: nonlinear dynamics

卡尔曼滤波算法

$$\begin{cases} \hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}(Y_{k+1} - C\bar{X}_{k+1}) \\ \bar{X}_{k+1} = A\hat{X}_k + Bu_k \\ K_{k+1} = (\bar{P}_{k+1}C^T)(C\ \bar{P}_{k+1}C^T + R\ )^{-1} \\ P_{k+1} = \bar{P}_{k+1} - K_{k+1}C\ \bar{P}_{k+1} \\ \bar{P}_{k+1} = AP_kA^T + Q \end{cases}$$

Parameters design:

$$\begin{cases} Q_k = E\left(w_k w_k^T\right) \\ R_{k+1} = E\left(n_{k+1} n_{k+1}^T\right) > 0 \\ P_0 = E\left(X_0 - EX_0\right)\left(X_0 - EX_0\right)^T \\ \bar{X}_0 = EX_0 \end{cases}$$

Require the exact information of :

linear model

noise's model

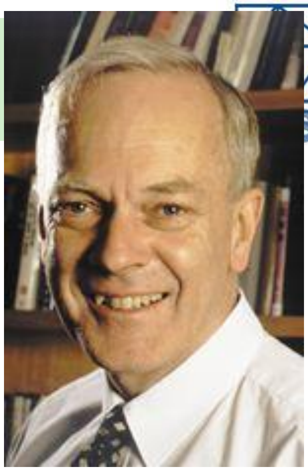
initial value's model

线性系统的卡尔曼滤波特性

$$\begin{cases} X_{k+1} = \bar{A}_k X_k + w_k \\ Y_{k+1} = \bar{C}_{k+1} X_{k+1} + n_{k+1} \end{cases}$$

$$k = 0, 1, 2, \ldots$$

$$\begin{cases} \hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}(Y_{k+1} - \bar{C}_{k+1}\bar{X}_{k+1}) \\ \bar{X}_{k+1} = \bar{A}_k \hat{X}_k \\ K_{k+1} = (\bar{P}_{k+1}\bar{C}_{k+1}^T)(\bar{C}_{k+1}\bar{P}_{k+1}\bar{C}_{k+1}^T + R_{k+1})^{-1} \\ P_{k+1} = \bar{P}_{k+1} - K_{k+1}\bar{C}_{k+1}\bar{P}_{k+1} \\ \bar{P}_{k+1} = \bar{A}_k P_k \bar{A}_k^T + Q_k \end{cases}$$



B. D. O. Anderson, ANU

特性分析

- **最优性：线性最小方差估计**

$$\hat{X}_k = \underset{Z_k \in \{\text{linear function of } Y_i, 1 \le i \le k\}}{\arg\min} E\left\{(Z_k - X_k)(Z_k - X_k)^T\right\}$$

- **稳定性：一致能观性 + $\{R_k, Q_k\}$具有上下界**

$$\sup_k \left\| E\left\{\left(\hat{X}_k - X_k\right)\left(\hat{X}_k - X_k\right)^T\right\} \right\| < \infty$$

- **一致性：可提供估计误差方差阵的（上）界**

$$E\left\{\left(\hat{X}_k - X_k\right)\left(\hat{X}_k - X_k\right)^T\right\} = P_k$$

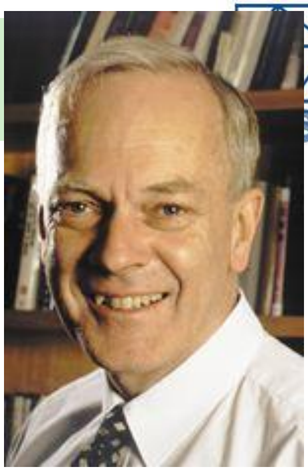
B. D. O. Anderson, etc., Detectability and Stabilizability of Time-Varying Discrete-Time Linear Systems, 1981.

线性系统的卡尔曼滤波特性

$$\begin{cases} X_{k+1} = \bar{A}_k X_k + w_k \\ Y_{k+1} = \bar{C}_{k+1} X_{k+1} + n_{k+1} \end{cases}$$

$$k = 0, 1, 2, \ldots$$

$$\begin{cases} \hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}(Y_{k+1} - \bar{C}_{k+1}\bar{X}_{k+1}) \\ \bar{X}_{k+1} = \bar{A}_k \hat{X}_k \\ K_{k+1} = (\bar{P}_{k+1}\bar{C}_{k+1}^T)(\bar{C}_{k+1}\bar{P}_{k+1}\bar{C}_{k+1}^T + R_{k+1})^{-1} \\ P_{k+1} = \bar{P}_{k+1} - K_{k+1}\bar{C}_{k+1}\bar{P}_{k+1} \\ \bar{P}_{k+1} = \bar{A}_k P_k \bar{A}_k^T + Q_k \end{cases}$$



B. D. O. Anderson, ANU

特性分析

- **最优性：线性最小方差估计**

$$\hat{X}_k = \underset{Z_k \in \{\text{linear function of } Y_i, 1 \le i \le k\}}{\arg\min} E\left\{(Z_k - X_k)(Z_k - X_k)^T\right\}$$

- **稳定性：一致能观性 ＋ $\{R_k, Q_k\}$具有上下界**

$$\sup_k \left\| E\left\{\left(\hat{X}_k - X_k\right)\left(\hat{X}_k - X_k\right)^T\right\}\right\| < \infty$$

- **一致性：可提供估计误差方差阵的（上）界**

$$E\left\{\left(\hat{X}_k - X_k\right)\left(\hat{X}_k - X_k\right)^T\right\} = P_k$$

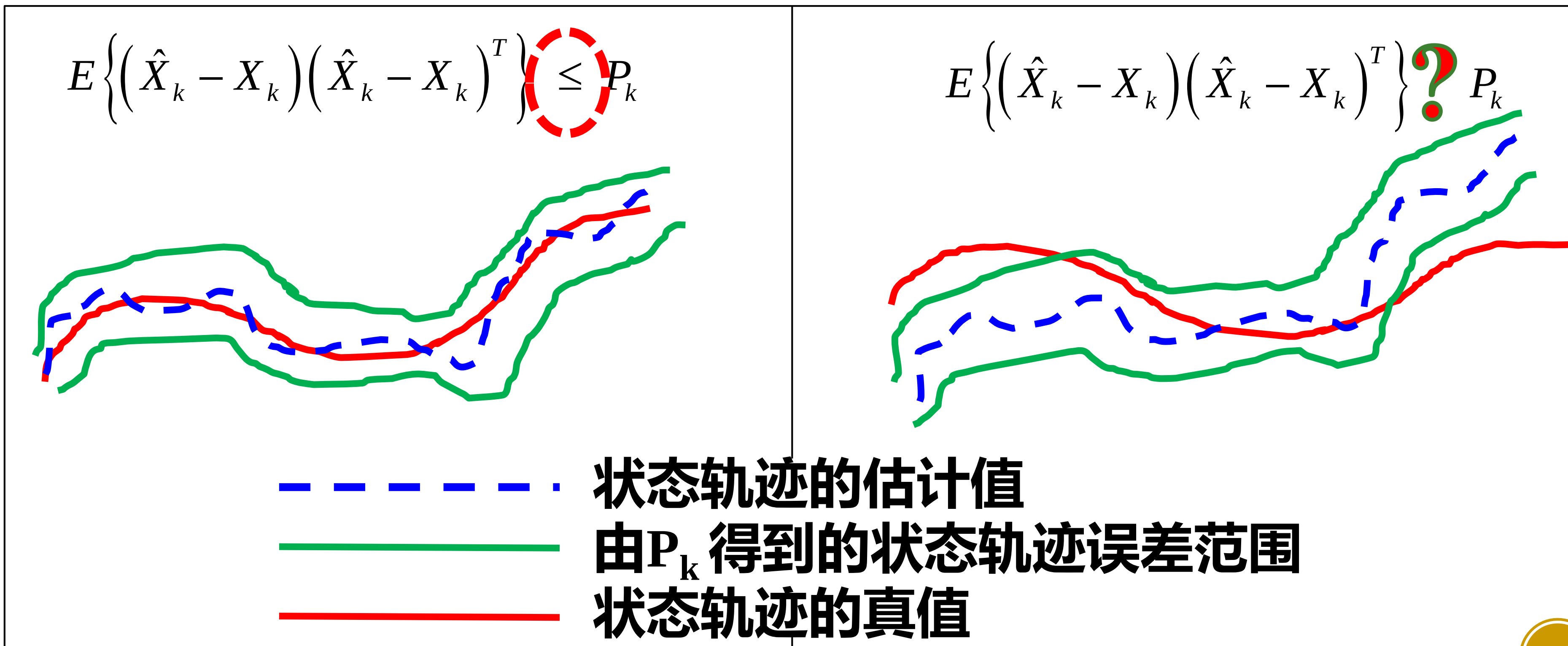
B. D. O. Anderson, etc., Detectability and Stabilizability of Time-Varying Discrete-Time Linear Systems, 1981.

一致性(Consistency)的意义

Definition Suppose X_k is a random vector. Let $\hat{X}_k$ and P_k be the estimate of X_k and the estimate of the corresponding error covariance matrix. Then **the pair $(\hat{X}_k, P_k)$ is said to be consistent (or of consistency)** if

$$E\{(\hat{X}_k - X_k)(\hat{X}_k - X_k)^T\} \leq P_k.$$

Julier, S. J., & Uhlmann, J. K. (1997).
A non-divergent estimation algorithm in the presence of unknown correlations.



非线性系统的滤波问题：

$$\begin{cases} X_{k+1} = f(X_k) + w_k \\ Y_{k+1} = g(X_{k+1}) + v_{k+1} \end{cases}$$

$k = 0, 1, 2, \ldots$

X_k : state, Y_k : measured output,

$(f(\bullet), g(\bullet))$: known C^1 functions

$(\{n_k\}, \{w_k\})$: uncorrelated zero-mean stochastic noise

Objective: linear minimum variance estimation, i.e.,

$$\hat{X}_k = \underset{Z_k \in \{\text{linear function of } Y_i, 1 \le i \le k\}}{\arg\min} E\left\{(Z_k - X_k)(Z_k - X_k)^T\right\}$$

对付非线性的主要方法：线性化

线性化
linearization

$$+$$

- **Kalman Filter, Kalman.**
- **H-infinity Filter, Banavar, etc.**
- **Set-valued Filter, Zhu, etc.**
- **…**

扩展卡尔曼滤波

$$\begin{cases} X_{k+1} = f(X_k) + w_k \\ Y_{k+1} = g(X_{k+1}) + v_{k+1} \end{cases}$$

$k = 0, 1, 2, \ldots$

X_k : state, Y_k : measured output,

$(f(\bullet), g(\bullet))$: known C^1 functions

$(\{v_k\}, \{w_k\})$: uncorrelated zero-mean

$$A_k = \left.\frac{\partial f}{\partial X}\right|_{\hat{X}_k},$$

第k+1步：

在滤波值点的线性部分；

$$C_{k+1} = \left.\frac{\partial g}{\partial X}\right|_{\bar{X}_{k+1}}$$

第k+1步：

在预测值点的线性部分

扩展卡尔曼滤波

$$\begin{cases} X_{k+1} = f(X_k) + w_k \\ Y_{k+1} = g(X_{k+1}) + v_{k+1} \end{cases}$$

$k = 0,1,2,\dots$

X_k : state, Y_k : measured output,

$(f(\bullet), g(\bullet))$: known C^1 functions

$(\{v_k\},\{w_k\})$: uncorrelated zero-mean

$$\begin{cases} \hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}\left(Y_{k+1} - g\left(\bar{X}_{k+1}\right)\right) \\ \bar{X}_{k+1} = F\left(\hat{X}_k, k\right) \\ K_{k+1} = (\bar{P}_{k+1}\bar{C}_{k+1}^T)(\bar{C}_{k+1}\bar{P}_{k+1}\bar{C}_{k+1}^T + R_{k+1})^{-1} \\ P_{k+1} = \bar{P}_{k+1} - K_{k+1}\bar{C}_{k+1}\bar{P}_{k+1} \\ \bar{P}_{k+1} = \bar{A}_k P_k \bar{A}_k^T + Q_k \end{cases}$$

$$A_k = \left.\frac{\partial f}{\partial X}\right|_{\hat{X}_k}, C_{k+1} = \left.\frac{\partial g}{\partial X}\right|_{\bar{X}_{k+1}}$$

Parameters design:

$$\begin{cases} Q_k = E\left(w_k w_k^T\right) \\ R_{k+1} = E\left(n_{k+1} n_{k+1}^T\right) > 0 \\ P_0 = E\left(X_0 - EX_0\right)\left(X_0 - EX_0\right)^T \\ \bar{X}_0 = EX_0 \end{cases}$$

Utilizing the linear part at the point of estimation value

$$\begin{cases} X_{k+1} = f(X_k) + w_k \\ Y_{k+1} = g(X_{k+1}) + v_{k+1} \end{cases}$$

$$k = 0, 1, 2, \ldots$$

Reif K., et al., Stochastic stability of the discrete-time extended Kalman filter, IEEE TAC, 1999

$$\begin{cases} \hat{X}_{k+1} = \bar{X}_{k+1} + K_{k+1}\left(Y_{k+1} - g\left(\bar{X}_{k+1}\right)\right) \\ \bar{X}_{k+1} = F\left(\hat{X}_k, k\right) \\ K_{k+1} = (\bar{P}_{k+1}\bar{C}_{k+1}^T)(\bar{C}_{k+1}\bar{P}_{k+1}\bar{C}_{k+1}^T + R_{k+1})^{-1} \\ P_{k+1} = \bar{P}_{k+1} - K_{k+1}\bar{C}_{k+1}\bar{P}_{k+1} \\ \bar{P}_{k+1} = \bar{A}_k P_k \bar{A}_k^{\,T} + Q_k, \quad A_k = \left.\frac{\partial f}{\partial X}\right|_{\hat{X}_k}, C_{k+1} = \left.\frac{\partial g}{\partial X}\right|_{\bar{X}_{k+1}} \end{cases}$$

EKF算法的主要理论结果

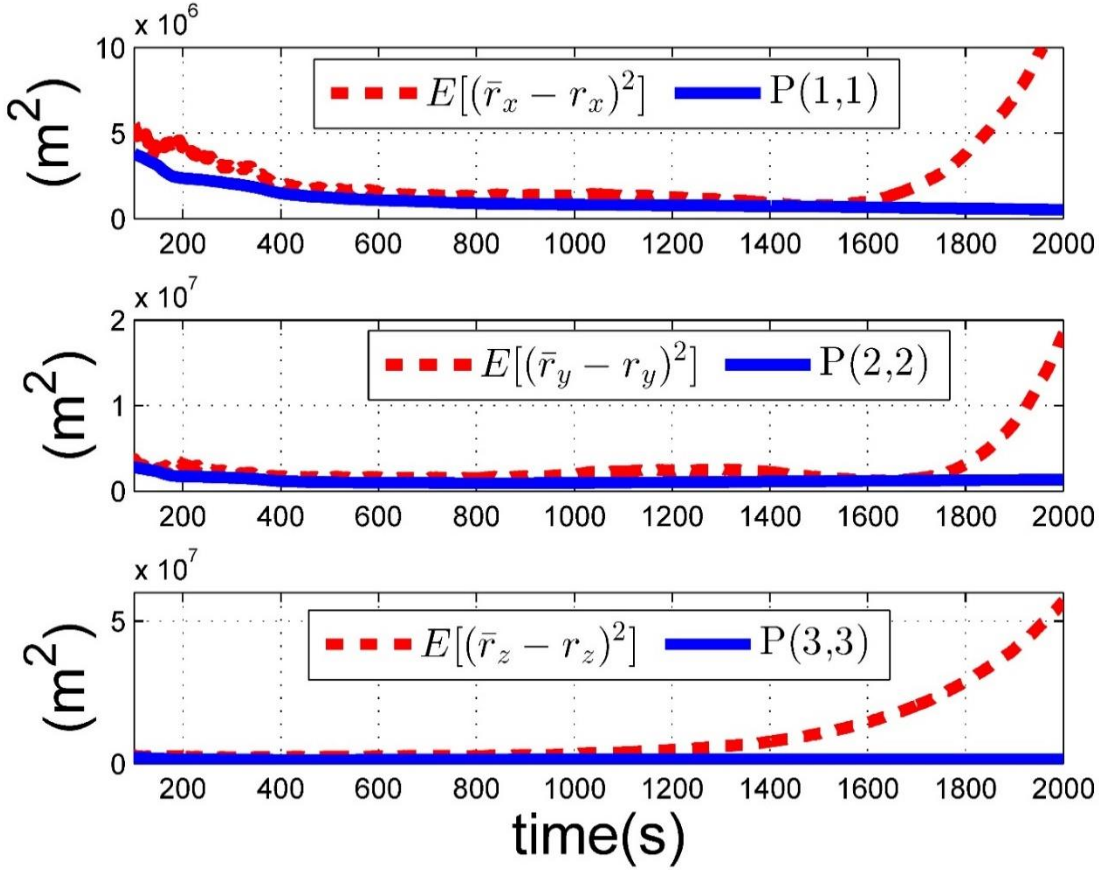
Conditions:
- **Exact information of nonlinear model**
- **Sufficiently small noise and initial error**
- **Uniformly observable condition**

稳定性: $\sup_k \left\| E\left\{\left(\hat{X}_k - X_k\right)\left(\hat{X}_k - X_k\right)^T\right\}\right\| < \infty$

最优性? **一致性?**

空间目标的雷达定轨

基于传统EKF的定轨算法：不能确保一致性



不能提供估计精度评估 ⇨ 可能产生错误决策

传统EKF：不能保证一致性的原因

$$E\left\{\left(\hat{X}_{k+1}-X_{k+1}\right)\left(\hat{X}_{k+1}-X_{k+1}\right)^T\right\}=P_{k+1}+\Delta P_{k+1}$$

线性化误差、估计误差、噪声的交叉项等

$$\begin{aligned}
\Delta P_{k+1} =& E[(I-K_kC_{k+1}^*)(A_k^*(\hat{X}_k-X_k)(\hat{X}_k-X_k)^TA_k^{*T})(I-K_kC_{k+1}^*)^T] \\
&-(I-K_kC_{k+1})A_kE[(\hat{X}_k-X_k)(\hat{X}_k-X_k)^T]{A_k}^T(I-K_kC_{k+1})^T \\
&+E[(I-K_kC_{k+1}^*)w_kw_k^T(I-K_kC_{k+1}^*)^T]-(I-K_kC_{k+1})E[w_kw_k^T](I-K_kC_{k+1})^T \\
&+E[K_k\psi_{k+1}^*(\cdot)\psi_{k+1}^{*T}(\cdot)K_k^T]+E[K_kv_{k+1}v_{k+1}^T{K_k}^T]-K_kE[v_{k+1}v_{k+1}^T]{K_k}^T \\
&+E[(I-K_kC_{k+1}^*)(A_k^*(\hat{X}_k-X_k)\varphi_k^{*T}(\cdot)+\varphi_k^*(\cdot)(\hat{X}_k-X_k)^TA_k^{*T})(I-K_kC_{k+1}^*)^T] \\
&+E[(I-K_kC_{k+1}^*)(\varphi_k^*(\cdot)\varphi_k^{*T}(\cdot))(I-K_kC_{k+1}^*)^T] \\
&-E[(I-K_kC_{k+1}^*)(A_k^*(\hat{X}_k-X_k)+\varphi_k^*(\cdot))\psi_{k+1}^{*T}(\cdot)K_k^T] \\
&-E[K_k\psi_{k+1}^*(\cdot)(A_k^*(\hat{X}_k-X_k)+\varphi_k^*(\cdot))^T(I-K_kC_{k+1}^*)^T].
\end{aligned}$$

where

$$\varphi_k^* = f(\hat{X}_k)-f(X_k)-A_k^*(\hat{X}_k-X_k),$$

$$\psi_{k+1}^* = g(\bar{X}_{k+1})-g(X_{k+1})-C_{k+1}^*(\bar{X}_{k+1}-X_{k+1})$$

一致性卡尔曼滤波 (CKF) 的基本思想

$$\begin{cases} \bar{X}_{k+1} = f(\hat{X}_k), \\ \bar{P}_{k+1} = A_k P_k {A_k}^T + Q_k + \Delta Q_k, \\ K_k = \bar{P}_{k+1} {C_{k+1}}^T (C_{k+1} \bar{P}_{k+1} {C_{k+1}}^T + R_{k+1})^{-1}, \\ \hat{X}_{k+1} = \bar{X}_{k+1} + K_k (y_{k+1} - g(\bar{X}_{k+1})), \\ P_{k+1} = (I - K_k C_{k+1}) \bar{P}_{k+1} (I - K_k C_{k+1})^T + K_k R_{k+1} {K_k}^T + \Delta R_{k+1}, \end{cases}$$

引入新的参数矩阵补偿 ΔP_k

where $\Delta Q_k \geq E[A_k^*(\hat{X}_k - X_k){\varphi_k}^T] + E[\varphi_k(\hat{X}_k - X_k)^T {A_k^*}^T] + E[\varphi_k {\varphi_k}^T]$

$$+ E[A_k^*(\hat{X}_k - X_k)(\hat{X}_k - X_k)^T {A_k^*}^T] - A_k E[(\hat{X}_k - X_k)(\hat{X}_k - X_k)^T]{A_k}^T,$$

$$\begin{aligned} \Delta R_{k+1} \geq & - E[(I - K_k C_{k+1}^*)(\bar{X}_{k+1} - X_{k+1}){\psi_{k+1}}^T {K_k}^T] \\ & - E[K_k \psi_{k+1}(\bar{X}_{k+1} - X_{k+1})^T (I - K_k C_{k+1}^*)^T] \\ & + E[K_k \psi_{k+1} {\psi_{k+1}}^T K_k^T] + E[K_k v_{k+1} v_{k+1}^T {K_k}^T] - K_k E[v_{k+1} v_{k+1}^T] {K_k}^T \\ & - (I - K_k C_{k+1}) E[(\bar{X}_{k+1} - X_{k+1})(\bar{X}_{k+1} - X_{k+1})^T](I - K_k C_{k+1})^T \\ & + E[(I - K_k C_{k+1}^*)(\bar{X}_{k+1} - X_{k+1})(\bar{X}_{k+1} - X_{k+1})^T (I - K_k C_{k+1}^*)^T]. \end{aligned}$$

Jiang, Y., Huang, Y., Xue, W.*, etc., On designing consistent extended Kalman filter. J Syst Sci Complex, 2017

空间目标运动模型

$$\begin{bmatrix}\dot{r}_x\\ \dot{r}_y\\ \dot{r}_z\\ \dot{v}_x\\ \dot{v}_y\\ \dot{v}_z\end{bmatrix}=\begin{bmatrix}v_x\\ v_x\\ v_x\\ -\frac{u_e}{q^3}r_x+2\omega v_y+\omega^2 r_x\\ -\frac{u_e}{q^3}r_y-2\omega v_x+\omega^2 r_y\\ -\frac{u_e}{q^3}r_z\end{bmatrix}\qquad q=\sqrt{r_x^2+r_y^2+r_z^2}$$

目标状态量测模型

$$y_k=h(\begin{bmatrix}r_{x,k}\\ r_{y,k}\\ r_{z,k}\end{bmatrix}-S_k)+v_k,\qquad h\left(\begin{bmatrix}z_x\\ z_y\\ z_z\end{bmatrix}\right)=\begin{bmatrix}\sqrt{z_x^2+z_y^2+z_z^2}\\ \arctan(\frac{z_y}{z_x})\\ \arctan(\frac{z_z}{\sqrt{z_x^2+z_y^2}})\end{bmatrix}$$

定理：对于空间目标定轨模型，如下参数设计

$$\Delta Q_k=\alpha_b A_{b,k}P_k A_{b,k}^T+6(1+\frac{1}{\alpha_b})\Delta Q_{\varphi,k}$$

$\alpha_b=0.01,$

$\Delta Q_{\varphi,k}=[(\sqrt{\bar{d}_{\varphi^2}(1)}+\bar{d}_{b,k}(1))^2,\cdots,(\sqrt{\bar{d}_{\varphi^2}(6)}+\bar{d}_{b,k}(6))^2],$

$\bar{d}_{\varphi^2}(1)=\bar{d}_{\varphi^2}(2)=\bar{d}_{\varphi^2}(3)=0,$

$\bar{d}_{\varphi^2}(i)=\frac{45}{2}trace\{\sqrt{P_{rr,k}}\frac{\partial^2 f_{b,i}}{\partial X_{r,b}}(\hat{X}_{r,bk})P_{rr,k}\frac{\partial^2 f_{b,i}}{\partial X_{r,b}}(\hat{X}_{r,bk})\sqrt{P_{rr,k}}^T\},$

$i=4,5,6,$ $\hat{X}_{r,b}$为$X_{r,b}$对应的滤波值，

$P_{rr,k}=\sqrt{P_{rr,k}}^T\sqrt{P_{rr,k}},$ $P_{rr,k+1}\in R^{3\times 3}$为$P_{k+1}$中位置矢量对应的对角块.

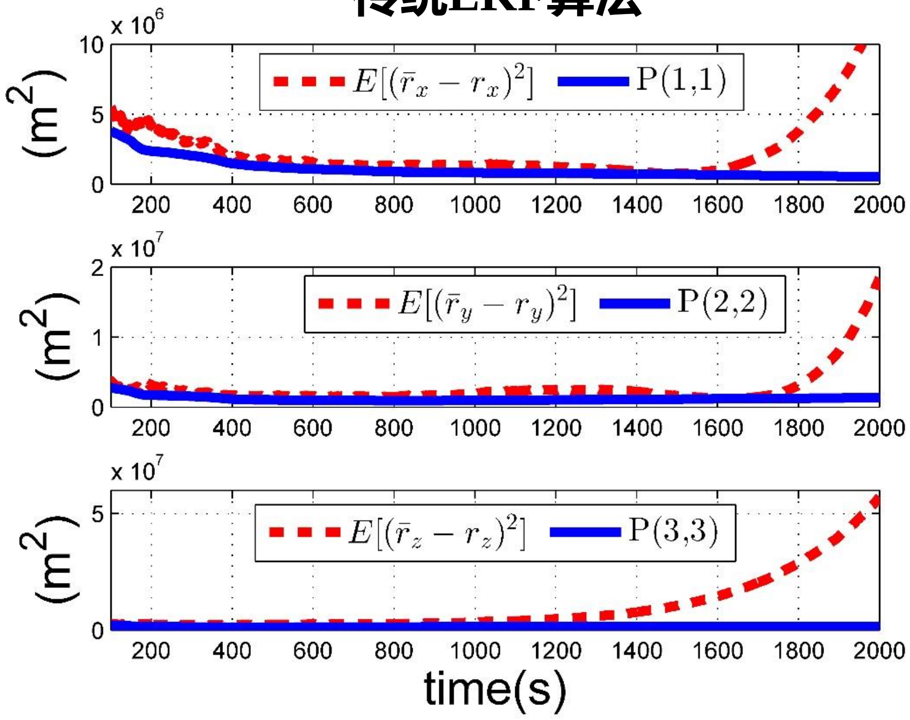
可得到CEKF算法，即满足一致性.

Jiang, Y., Huang, Y., Xue, W.*, etc., On designing consistent extended Kalman filter. J Syst Sci Complex, 2017

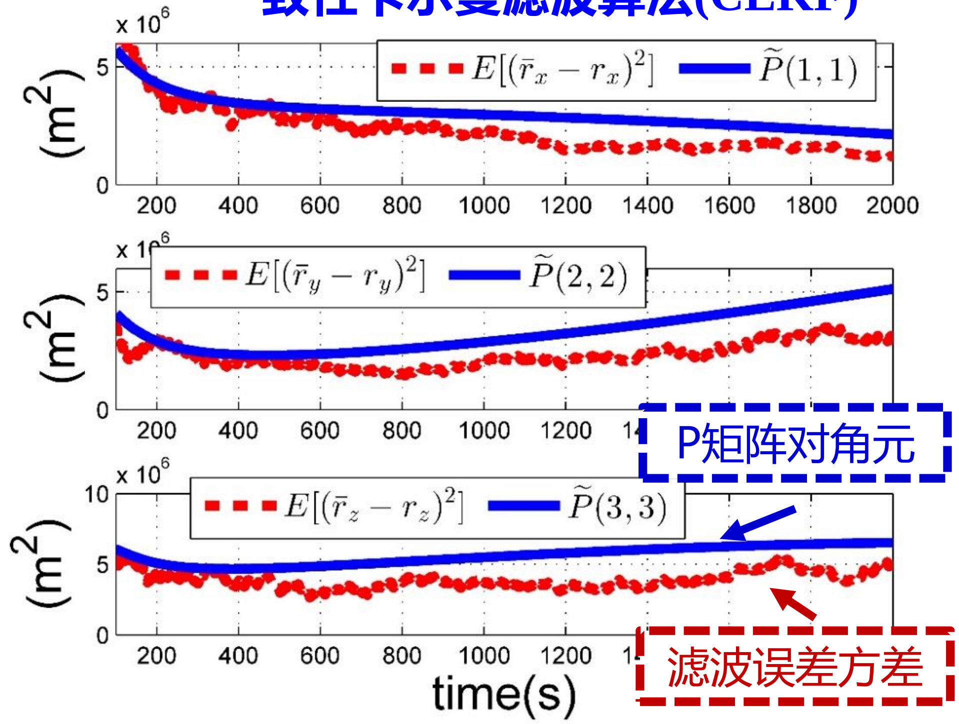
空间目标定轨

仿真结果对比分析

传统EKF算法

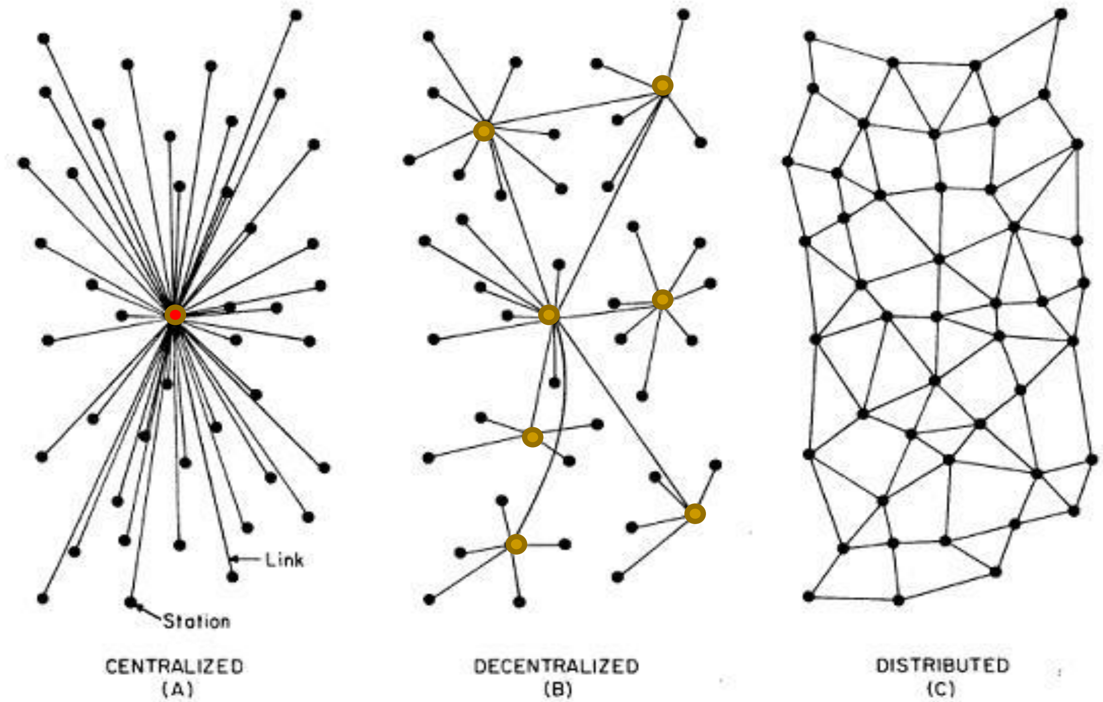


一致性卡尔曼滤波算法(CEKF)



传感器网络的卡尔曼滤波算法设计

传感器网络通讯拓扑网络：



Strategy	Center	Vital Sensors	Information Exchange
中心化(Centralized)	Yes	One (Highest)	With Center
分散化(Decentralized)	No	Several (Middle)	With Neighbors
分布化(Distributed)	No	No	With Neighbors

降低通讯带宽、增强估计鲁棒，容易并行计算

拓扑图的一些基本定义

- **网络拓扑:**
  - **固定有向图** $\mathcal{G}=(\mathcal{V},\mathcal{E},\mathcal{H}),$
  - **节点集合** $\mathcal{V}=\{1,2,\cdots,N\}$
  - **边集合** $\mathcal{E}\subseteq\mathcal{V}\times\mathcal{V}$
  - **加权邻接矩阵** $\mathcal{H}=[a_{ij}]$

- **传感器 *i* 的邻居集合:** $\{j\in\mathcal{V}\,|\,(i,j)\in\mathcal{E}\}\bigcup\{i\}\triangleq\mathcal{N}_i$

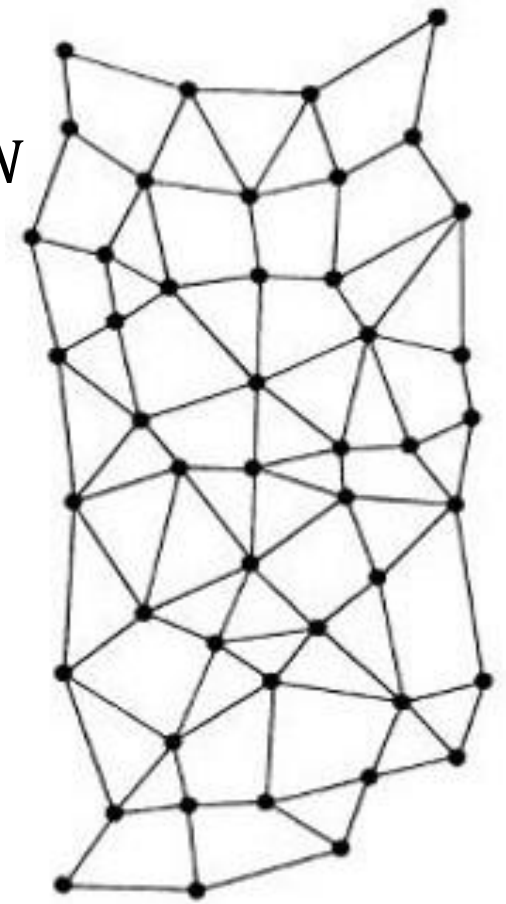
- **有向图：每一条边存在指向，即** $(i,j)\neq(j,i)$

- **强连接有向图: 任意一对节点之间都存在一条有向路**

传感器网络下的分布式估计问题模型

系统模型

$$\begin{cases} x_{k+1} = A_k x_k + \omega_k, \\ y_{k,i} = H_{k,i} x_k + v_{k,i} \quad i = 1,2,\cdots,N \end{cases}$$



x_k: 系统状态

A_k: 系统矩阵

$y_{k,i}$: 第i传感器节点的量测量

$H_{k,i}$: 第i传感器节点的量测矩阵

$\left(\{w_k\},\{v_{k,i}\}\right)$: 不相关零均值

传感器网络下的滤波算法研究

- **分布式滤波算法的结构设计** **(融合机制与局部滤波)**

  Speranzon et al (IEEE JSAC,2008), Cattivelli and Sayed (IEEE TAC,2010), Hu et al (IEEE TSP,2011), Battistelli and Chisci (Automatica,2016)

- **含不确定动态系统的滤波估计算法设计** **(鲁棒估计与滤波)**

  Chen et al (IEEE TAES,2015), Huang et al (IEEE TAC, 2020)

- **针对特定量测的滤波算法设计** **(角度量测，距离量测)**

  Mohammadi and Asif(IEEE TSP,2014), Lin et al(Automatica,2017)

- **针对复杂网络的滤波算法设计** **(量测缺失、连接中断)**

  Hu et al (IEEE Trans. Syst. Man Cybern. Syst. 2020), Liu and Cheng (IEEE TCNS,2022)

- **……**

传感器网络下的分布式卡尔曼型滤波框架

系统模型

$$\begin{cases} x_{k+1} = A_k x_k + \omega_k, \\ y_{k,i} = H_{k,i} x_k + v_{k,i} \quad i = 1,2,\cdots,N \end{cases}$$

分布式滤波

$$\begin{cases} \phi_{k,i} = \bar{x}_{k,i} + K_{k,i}(y_{k,i} - H_{k,i}\bar{x}_{k,i}), \\ \hat{x}_{k,i} = \sum_{j\in\mathcal{N}_i} W_{k,i,j}\phi_{k,j}, \\ \bar{x}_{k+1,i} = A_k \hat{x}_{k,i}. \end{cases}$$

量测更新　　最优增益

局部融合　　最优加权

预报

Covariance Intersection融合方式 (Arambel P O et.al, 2001 ACC)

$$W_{k,i,j} = \left(\sum_{j\in\mathcal{N}_i} w_{k,i,j}\tilde{P}_{k,j}^{-1}\right)^{-1} w_{k,i,j}\tilde{P}_{k,j}^{-1}, \quad \sum_{j\in\mathcal{N}_i} w_{k,i,j} = 1, w_{k,i,j} \geq 0,$$

邻居节点提供的$\tilde{P}$阵越大，对该邻居的估计值加权越小

传感器网络下的分布式卡尔曼型滤波框架

分布式滤波

$$\begin{cases} \phi_{k,i} = \bar{x}_{k,i} + K_{k,i}(y_{k,i} - H_{k,i}\bar{x}_{k,i}), \\ \hat{x}_{k,i} = \sum_{j \in \mathcal{N}_i} W_{k,i,j}\phi_{k,j}, \\ \bar{x}_{k+1,i} = A_k \hat{x}_{k,i}. \end{cases}$$

→ 量测更新(最优增益)

→ 局部融合(最优加权)

→ 预报

- S. Das, A. Jadbabaie and J. M. F. Moura (2015 IEEE TSP, 2014 Automatica)
- F. S. Cattivelli and A. H. Sayed (2010 IEEE TAC, 2008 IEEE TSP)
- G. Battistelli and L. Chisci (2015 IEEE TAC, 2014 Automatica)
- H. Dong, Z. Wang, H. Gao (2013 IEEE TIE)
- Y. Huang, Y. Zhang, P. Shi and J. Chambers, (2020 IEEE TAC)
- D. Su, H. Kong, S. Sukkarieh and S. Huang (2021 IEEE TOR)
- …

分布式卡尔曼型滤波设计的关键问题

□ 滤波算法的稳定性（一致性满足时P阵的有界性）

- 每个传感器节点满足局部能观性时容易得到

  Cattivelli & Sayed（2010 IEEE TAC), Hu et al. (2011 IEEE TSP)

- 只满足全局能观性而局部不满足能观性的情况?

□ 各节点滤波算法的一致性

- 线性已知模型、噪声模型已知时已证明（经典KF理论）
- 系统含未知动态、噪声模型不确定性的情况?

一致性分布式卡尔曼滤波(CDKF)

Assumption 1 $\{\omega_k\}_{k=0}^{\infty}$ *and* $\{v_{k,i}\}_{k=0}^{\infty}$ *are zero-mean, Gaussian, white and uncorrelated. Also,* $R_{k,i}$ *is positive definite,* $\forall k \geq 0$. *There exist two constant positive definite matrices* $\bar{Q}_1$ *and* $\bar{Q}_2$ *such that* $\bar{Q}_1 \leq Q_k \leq \bar{Q}_2, \forall k \geq 0$. *The initial state* x_0 *is generated by a zero-mean white Gaussian process independent of* $\{\omega_k\}_{k=0}^{\infty}$ *and* $\{v_{k,i}\}_{k=0}^{\infty}$, *subject to* $E\{x_0 x_0^T\} = P_0$.

□ 噪声协方差上界已知

Assumption 2 *The system (1) is uniformly completely observable, i.e., there exist a positive integer* $\bar{N}$ *and positive constants* α, β *such that for any* $k \geq 0$,

$$\begin{cases} 0 < \alpha I_n \leq \sum_{j=k}^{k+\bar{N}} \Phi_{j,k}^T H_j^T R_j^{-1} H_j \Phi_{j,k} \leq \beta I_n, \\ \Phi_{k,k} \triangleq I_n, \Phi_{k+1,k} \triangleq A_k, \Phi_{j,k} \triangleq \Phi_{j,j-1} \cdots \Phi_{k+1,k}, \\ H_k \triangleq blockcol\{H_{k,1}, H_{k,2}, \cdots, H_{k,N}\}, \\ R_k \triangleq blockdiag\{R_{k,1}, R_{k,2}, \cdots, R_{k,N}\}. \end{cases}$$

□ 全局能观：所有量测量集合在一起使得系统能观

□ 节点滤波设计：只能利用局部的量测模型

X. He, W. Xue*, and H. Fang, Consistent Distributed State Estimation with Global Observability Over Sensor Network, Automatica, 2018.

一致性分布式卡尔曼滤波(CDKF)

Assumption 3 *The topology of the network $\mathcal{G}=(\mathcal{V},\mathcal{E},\mathcal{A})$ is a fixed directed graph and it is strongly connected.*

Assumption 4 *There exists a positive scalar β_1, such that $\lambda_{max}(A_k A_k^T) \leq \beta_1, \forall k \geq 0$.*

Assumption 5 *There exist a sequence set $\mathcal{K}=\{k_l, l \geq 1\}$, an integer $L \geq N+\bar{N}$ and a scalar $\beta_2 > 0$, such that*

$$\begin{cases} \sup_{l\geq 1}(k_{l+1}-k_l) < \infty, \\ \inf_{l\geq 1}(k_{l+1}-k_l) > 0, \\ \lambda_{min}(A_{k_l+s}A_{k_l+s}^T) \geq \beta_2, \forall k_l \in \mathcal{K}, s=0,\cdots,L-1. \end{cases}$$

- 系统矩阵：可在多个时刻奇异
- 系统矩阵奇异带来的困难：难以保证矩阵P的有界

X. He, W. Xue*, and H. Fang, Consistent Distributed State Estimation with Global Observability Over Sensor Network, Automatica, 2018.

一致性分布式卡尔曼滤波(CDKF)

CDKF 算法

Prediction:

$$\bar{x}_{k,i} = A_{k-1}\hat{x}_{k-1,i},$$
$$\bar{P}_{k,i} = A_{k-1}P_{k-1,i}A_{k-1}^T + Q_{k-1},$$

Measurement Update:

$$\phi_{k,i} = \bar{x}_{k,i} + K_{k,i}(y_{k,i} - H_{k,i}\bar{x}_{k,i}),$$
$$K_{k,i} = \bar{P}_{k,i}H_{k,i}^T(H_{k,i}\bar{P}_{k,i}H_{k,i}^T + R_{k,i})^{-1},$$
$$\tilde{P}_{k,i} = (I - K_{k,i}H_{k,i})\bar{P}_{k,i},$$

Local Fusion: Receiving $(\phi_{k,j}, \tilde{P}_{k,j})$ from neighbors $j \in \mathcal{N}_i$

$$\hat{x}_{k,i} = P_{k,i}\sum_{j\in\mathcal{N}_i} w_{k,i,j}\tilde{P}_{k,j}^{-1}\phi_{k,j},$$
$$P_{k,i} = (\sum_{j\in\mathcal{N}_i} w_{k,i,j}\tilde{P}_{k,j}^{-1})^{-1},$$

$\tilde{P}_{k,j}$:邻居 j 的估计精度

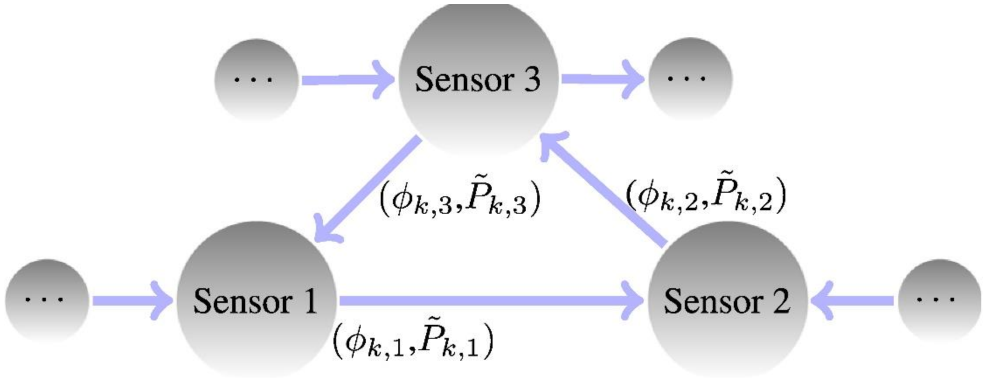
$w_{k,i,j}$: 为了得到更小$P_{k,i}$设计时变CI 权重

Design $w_{k,i,j}(\geq 0)$, such that $\sum_{j\in\mathcal{N}_i} w_{k,i,j} = 1.$

Initialization:

$$\hat{x}_{0,i} = 0, P_{0,i} \geq P_0.$$

CDKF算法的交流拓扑



X. He, W. Xue*, and H. Fang, Consistent Distributed State Estimation with Global Observability Over Sensor Network, Automatica, 2018.

一致性分布式卡尔曼滤波(CDKF)

理论结果

Theorem 3.1 Considering the system (1), under Assumption 1, the pair $(\hat{x}_{k,i}, P_{k,i})$ of the CDKF is consistent, i.e.,

Consistency: $$E\{(\hat{x}_{k,i} - x_k)(\hat{x}_{k,i} - x_k)^T\} \leq P_{k,i}, \forall i \in \mathcal{V}, k \geq 0. \quad (4)$$

Theorem 3.2 Under Assumptions 1–5, there exists a positive definite matrix $\hat{P}$, such that

Stability: $$P_{k,i} \leq \hat{P} < \infty, \quad \forall i \in \mathcal{V}, \forall k \geq 0, \quad (11)$$

where $P_{k,i}$ is the parameter matrix of the CDKF.

Theorem 3.3 Under Assumptions 1–3, if $\{A_k\}_{k=0}^{\infty}$ belongs to a nonsingular compact set, then there is

Unbiased: $$\lim_{k \to +\infty} E\{\hat{x}_{k,i} - x_k\} = 0, \quad \forall i \in \mathcal{V}. \quad (17)$$

X. He, W. Xue*, and H. Fang, Consistent Distributed State Estimation with Global Observability Over Sensor Network, Automatica, 2018.

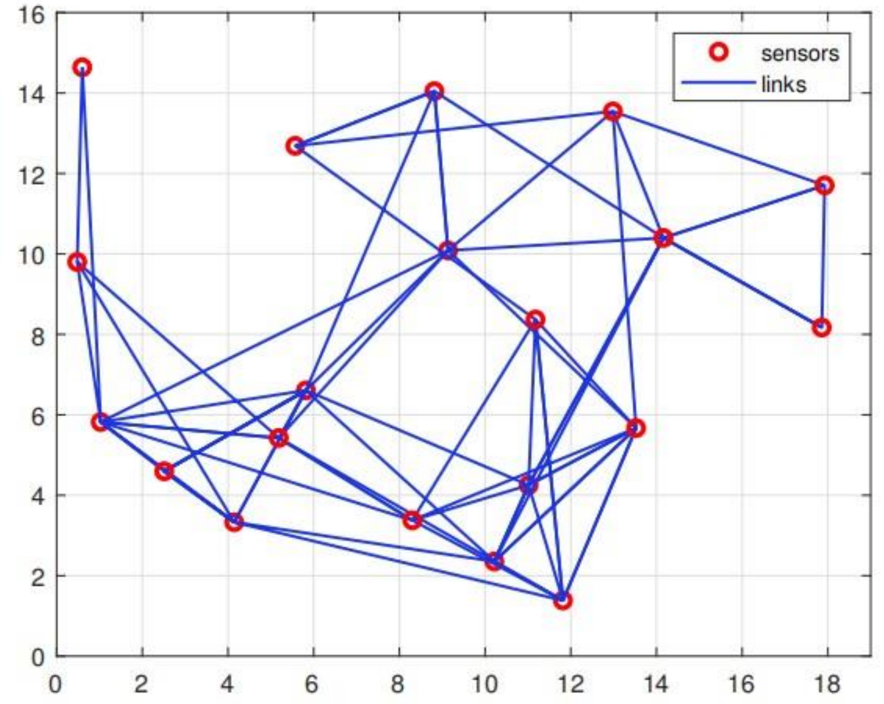
传感器网络的数值仿真结果

- **存在有界的不确定动态**
- **使用错配的协方差矩阵**

$$A_k = \begin{bmatrix} 1+0.3sin(0.1k) & 0 & 0 \\ 0.1 & cos(0.5) & 0.4+sin(0.5) \\ 0.1 & 0.4-sin(0.5) & cos(0.5) \end{bmatrix}$$

$$F_k(x_k) = \frac{1}{2k}\begin{bmatrix} sin(x_k(1)) \\ sin(x_k(2)) \\ sin(x_k(3)) \end{bmatrix} + 0.2\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$E[F_k(x_k)F_k(x_k)^T] \leq D_k = \begin{bmatrix} 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0.5 \end{bmatrix}$$



节点随机选择观测矩阵

$$H^{(1)} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, H^{(2)} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, H^{(3)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, H^{(4)} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

500 次蒙特卡洛实验:

$$ERROR_k(l) = \frac{1}{500}\sum_{j=1}^{500}(\hat{x}_{k,i}^{j}(l) - x_k(l))^2$$

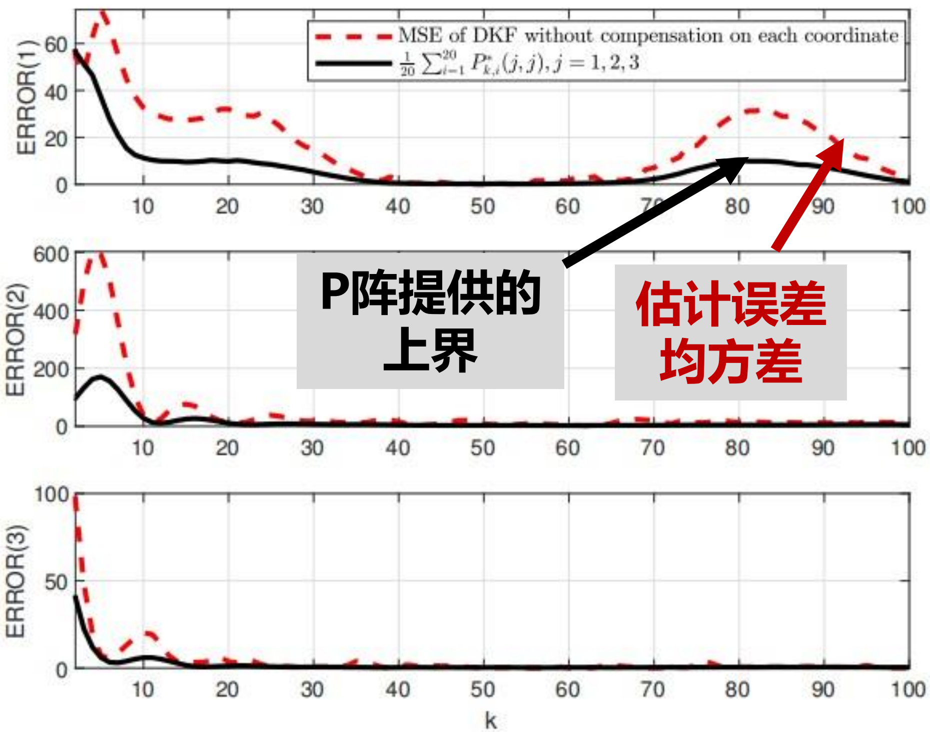
$0 \le \delta_k \le 4,\ \ 0 \le \alpha_{k,i} \le 4,\ \ 0 \le \beta_i \le 3$

$\bar{\delta}_k = 0,\ \ \bar{\alpha}_{k,i} = 0,\ \ \bar{\beta}_i = 0$

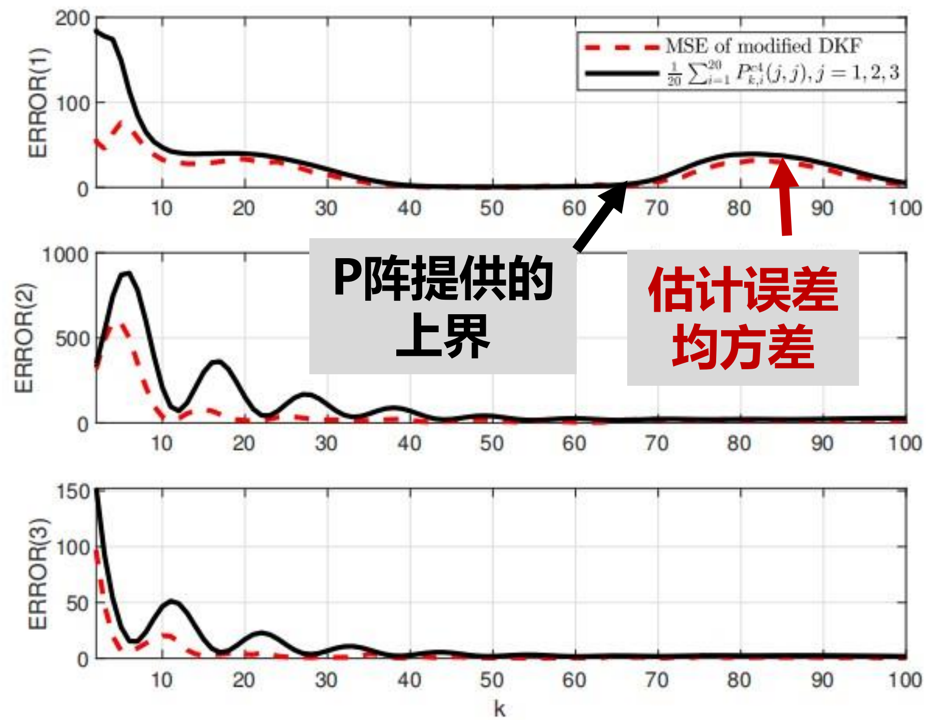
$\delta_k = 4,\ \ \alpha_{k,i} = 4,\ \ \beta_i = 3$

$4 \le \bar{\delta}_k \prec 5,\ \ 4 \le \bar{\alpha}_{k,i} \prec 5,\ \ 3 \le \bar{\beta}_i \prec 4$

传统卡尔曼滤波



一致性卡尔曼滤波



应用研究：基于集群的机动目标分布式定位

目标动态：

$$r_{k+1,p} = r_{k,p} + \Delta_k$$

目标未知机动

$$E\left[\Delta_k \Delta_k^T\right] \le \begin{bmatrix} (1000t_s)^2 & 0 & 0 \\ 0 & (1000t_s)^2 & 0 \\ 0 & 0 & (1000t_s)^2 \end{bmatrix}$$

飞行器-i 的量测量

$$\bar{y}_i(kt_s) = \begin{bmatrix} b_{m_i}(kt_s) \\ e_{m_i}(kt_s) \\ r_{m_i}(kt_s) \end{bmatrix}$$

$$w^i(kt_s) = r_k - r_{k,A_i},\ w^i = [w_x^i, w_y^i, w_z^i]^T$$

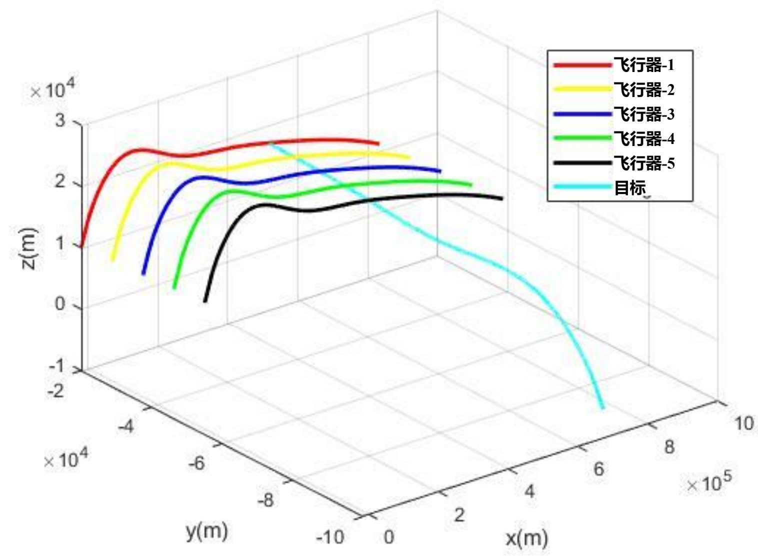
b_{m_i} :目标相对于传感器 i 的方位角量测

e_{m_i} :目标相对于传感器 i 的高低角量测

r_{m_i} :目标与传感器 i 之间距离量测

r_{k,A_i}： 传感器 i 的位置矢量

$n_{b_i}, n_{e_i}, n_{r_i}$： 量测噪声



应用研究：基于集群的机动目标分布式定位

位置量测的无偏转换

$$\tilde{r}_{k,p}^{(i)} \triangleq diag\left(\begin{bmatrix} E\left\{\cos(n_{e_i}(kt_s))\cos(n_{b_i}(kt_s))\right\} \\ E\left\{\cos(n_{e_i}(kt_s))\cos(n_{b_i}(kt_s))\right\} \\ E\left\{\cos(n_{e_i}(kt_s))\right\} \end{bmatrix}\right)^{-1} r_{m_i}(kt_s)\begin{bmatrix} \cos\left(e_{m_i}(kt_s)\right)\cos\left(b_{m_i}(kt_s)\right) \\ \cos\left(e_{m_i}(kt_s)\right)\sin\left(b_{m_i}(kt_s)\right) \\ \sin\left(e_{m_i}(kt_s)\right) \end{bmatrix} + r_{k,A_i}$$

融合卡尔曼滤波算法

局部滤波

$$\begin{cases} \bar{r}_{k+1,p}^{(i)} = \hat{r}_{k,p}^{(i)} \\ \bar{P}_{k+1}^{(i)} = P_k^{(i)} + Q_k \\ K_k^{(i)} = \bar{P}_{k+1}^{(i)}\left(\bar{P}_{k+1}^{(i)} + R_{k+1,p}^{(i)}\right)^{\dagger} \\ \phi_{k+1}^{(i)} = \bar{r}_{k+1,p}^{(i)} + K_k^{(i)}\left(\tilde{r}_{k+1,p}^{(i)} - \bar{r}_{k+1,p}^{(i)}\right) \\ \tilde{P}_{k+1}^{(i)} = \left(I_3 - K_k^{(i)}\right)\bar{P}_{k+1}^{(i)}\left(I_3 - K_k^{(i)}\right)^T + K_k^{(i)} R_{k+1,p}^{(i)}\left(K_k^{(i)}\right)^T \end{cases}$$

$$Q_k \geq E[\Delta_k\Delta_k^T] - E[(\hat{r}_{k,p}^{(i)} - r_{k,p})\Delta_k^T] - E[\Delta_k(\hat{r}_{k,p}^{(i)} - r_{k,p})^T]$$

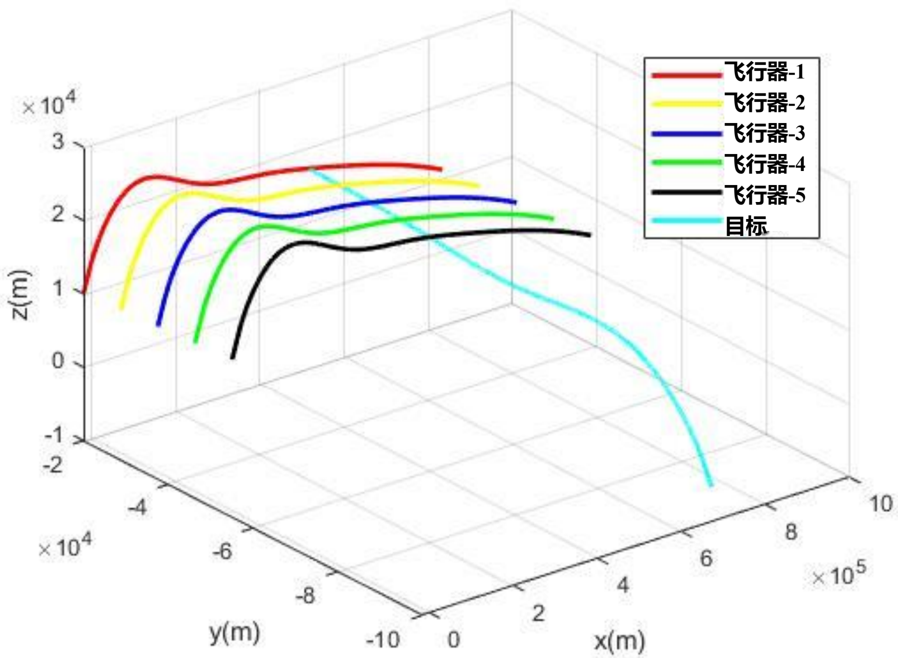
$$Q_k = 0.2P_k + 6\begin{bmatrix} (1000t_s)^2 & 0 & 0 \\ 0 & (1000t_s)^2 & 0 \\ 0 & 0 & (1000t_s)^2 \end{bmatrix}$$

局部融合

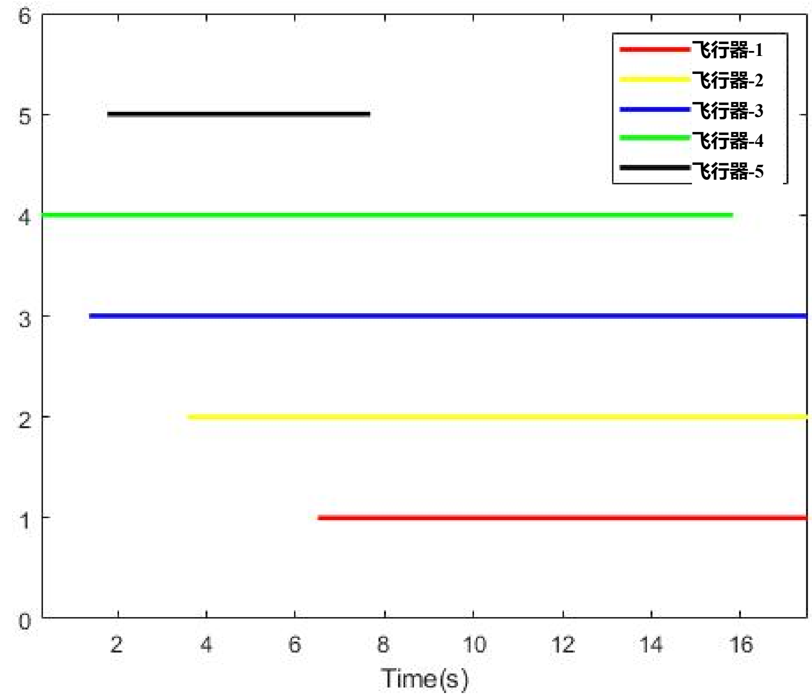
$$\begin{cases} \hat{r}_{k+1,p}^{(i)} = P_{k+1}^{(i)}\sum_{j\in N_i}\dfrac{1}{|N_i|}\left(\tilde{P}_{k+1}^{(j)}\right)^{-1}\phi_{k+1}^{(j)} \\ P_{k+1}^{(i)} = \left(\sum_{j\in N_i}\dfrac{1}{|N_i|}\left(\tilde{P}_{k+1}^{(j)}\right)^{-1}\right)^{-1} \end{cases}$$

应用研究：基于集群的机动目标分布式定位

各飞行器节点于目标的轨迹

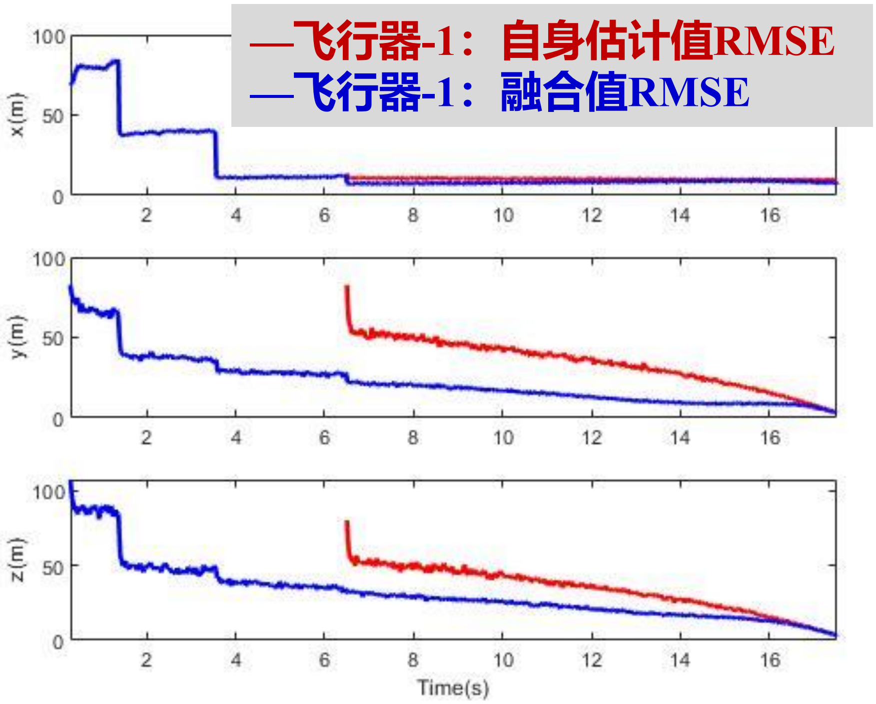


各飞行器产生目标位置量测的时间区间

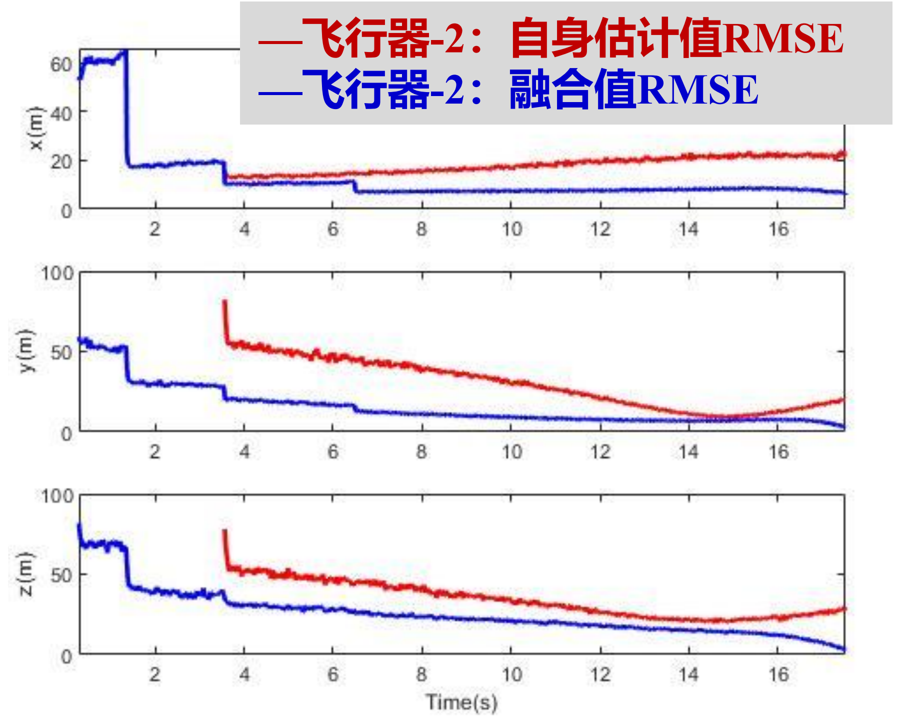


应用研究：基于集群的机动目标分布式定位

飞行器-1
局部估计值与融合估计值



飞行器-2
局部估计值与融合估计值

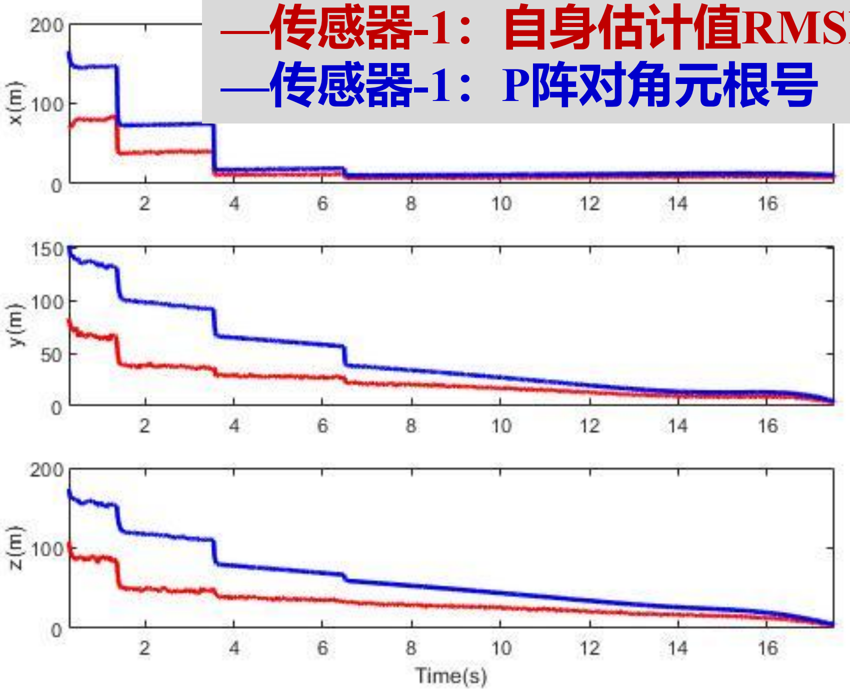


500次蒙特卡洛实验

应用研究：基于集群的机动目标分布式定位

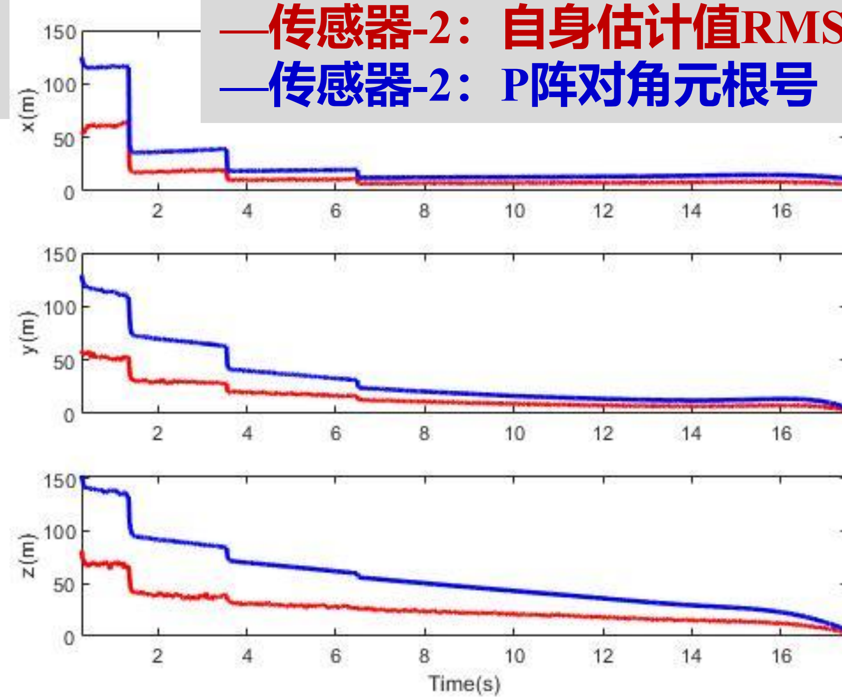
飞行器-1
估计误差及误差范围评估值

—传感器-1：自身估计值RMSE
—传感器-1：P阵对角元根号



飞行器-2
估计误差及误差范围评估值

—传感器-2：自身估计值RMSE
—传感器-2：P阵对角元根号



500次蒙特卡洛实验

$$RMSE_{k,i}(l)=\sqrt{\frac{1}{500}\sum_{j=1}^{500}\left(\hat{r}_{k,p}^{(i)}(l)-r_{k,p}(l)\right)^{2}},l=x,y,z$$

概要

- 研究背景和问题描述
- 卡尔曼滤波算法及其主要特性
- 卡尔曼滤波算法的一致性
- **扩张状态卡尔曼滤波算法**

机动目标定轨：未知的非线性动态（加速度）

Filtering problem of nonlinear uncertain systems:

$$\begin{cases} X_{k+1} = \bar{A}_k X_k + \bar{B}_k F(X_k, k) + w_k \\ Y_{k+1} = \bar{C}_{k+1} X_{k+1} + n_{k+1} \end{cases}$$

$$k = 0, 1, 2, \ldots$$

X_k : state, Y_k : measured output

$(\bar{A}_k, \bar{B}_k, \bar{C}_{k+1})$: known matrices

$F(X_k, k)$: nonlinear uncertain dynamics

$(\{n_k\}, \{w_k\})$: uncorrelated zero-mean stochastic noise

新问题

- 如何处理系统的非线性不确定动态？
- 如何确保滤波算法的稳定性等特性？

控制系统中存在着多种多样的不确定性

DARPA, HTV-2

Boston dynamics, LS3

ABB, YuMi

主要不确定性的种类	物理来源	数学描述
1. 未建模/未知的动态	复杂摩擦力、未建模动态等	非线性函数
2. 外部的扰动	未知的外部环境变化	时变信号
3. 噪声	传感器、执行器的误差	具有统计性质的随机变量

控制系统中存在着多种多样的不确定性

$$\begin{cases} X_{k+1} = \bar{A}_k X_k + \bar{B}_k F(X_k, k) + w_k \\ Y_{k+1} = \bar{C}_{k+1} X_{k+1} + n_{k+1} \end{cases}$$

$k = 0, 1, 2, \ldots$

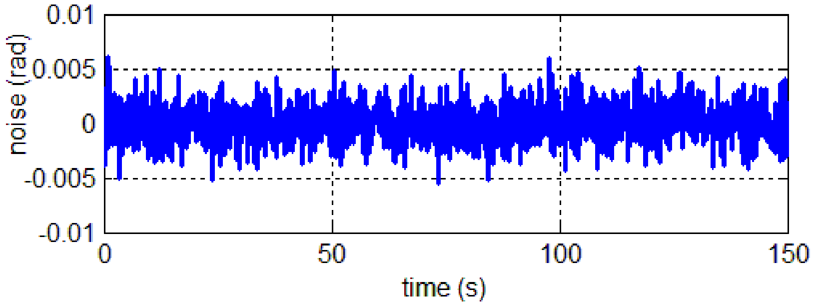
X_k : state, Y_k : measured output

$(\bar{A}_k, \bar{B}_k, \bar{C}_{k+1})$: known matrices

$F(X_k, k)$: nonlinear uncertain dynamics

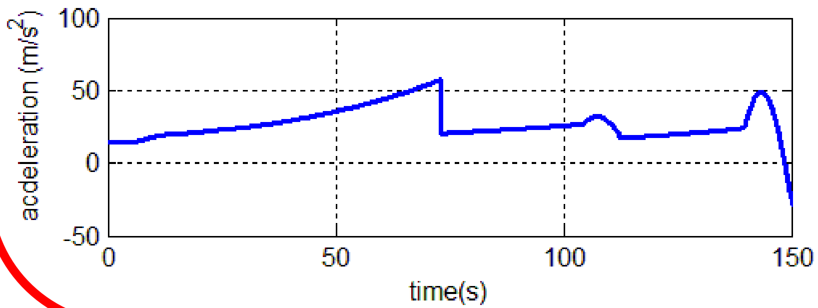
$(\{n_k\}, \{w_k\})$: uncorrelated zero-mean stochastic noise

机动目标动力学系统中的两种不确定性：



- **随机量测噪声**

高频信号：优化滤波增益减小其影响



- **非线性不确定动态**

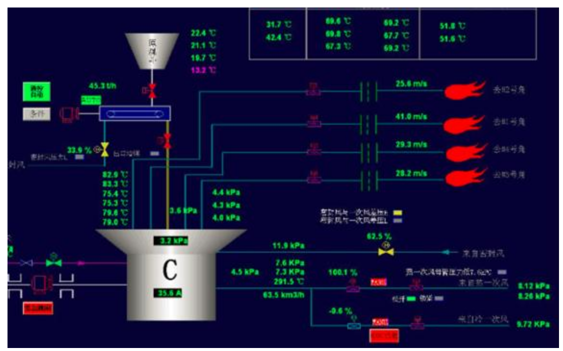
低频信号：实时估计并补偿

自抗扰控制(ADRC)的核心:
扩张状态观测器(Extended State Observer, ESO)

$$\begin{cases} X_{k+1} = \bar{A}_k X_k + \bar{B}_k F(X_k, k) + w_k \\ Y_{k+1} = \bar{C}_{k+1} X_{k+1} + n_{k+1} \end{cases} \Leftarrow \begin{bmatrix} \hat{X}_{k+1} \\ \hat{F}_{k+1} \end{bmatrix} = \begin{bmatrix} \bar{A}_k & \bar{B}_k \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{X}_k \\ \hat{F}_k \end{bmatrix} + \bar{L}_k (Y_k - \bar{C}_k \hat{X}_k)$$

- **将未知非线性动态视为一个低频的时变信号**
- **设计ESO对原始状态和未知非线性动态同时进行在线估计**

High energy particle Accelerators,
Provided by Zhiqiang Gao



Control system of thermal power generating unit,
Provided by Donghai Li

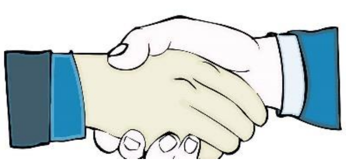
Texas Instrument New Motion Control Chips,
Provided by TI

J. Han. A class of extended observers for uncertain systems. Control and Decision, 10(1):85--88, 1995.

采用不同的方法对付不同类型的不确定

- Stochastic noise: attenuated by the technique of KF
- Nonlinear uncertain dynamics: estimated by the extended state method

	Gain tuning	Nonlinear uncertain dynamics
KF	Optimized by KF type algorithm	?
ESO	?	Being treated as an extended state



扩张状态卡尔曼滤波器(ESKF)

Filtering problem for nonlinear uncertain systems

$$\begin{cases} X_{k+1} = \bar{A}_k X_k + \bar{B}_k F(X_k, k) + w_k, \\ Y_k = \bar{C}_k X_k + n_k, \end{cases} \quad k = 0, 1, \ldots,$$

$$A_k = \begin{bmatrix} \bar{A}_k & \bar{B}_k \\ 0 & I \end{bmatrix}, \quad B_k = \begin{bmatrix} 0 \\ I \end{bmatrix}, \quad C_k = \begin{bmatrix} \bar{C}_k & 0 \end{bmatrix}$$

A1. (A_k, C_k) is uniformly observable. ***Observability***

A2. $\{w_k\}_0^\infty$ and $\{n_k\}_0^\infty$ are uncorrelated zero-mean Gaussian random sequences and

$$E(n_k n_k^T) \leqslant R_k, \quad E(w_k w_k^T) \leqslant S_k,$$

Measurement noise)

where $\{R_k\}_{k=0}^\infty$ and $\{S_k\}_{k=0}^\infty$ are positive and uniformly lower and upper bounded. In addition, $\{w_k\}_0^\infty$, $\{n_k\}_0^\infty$ and X_0 are mutually independent.

A3. ***Initial estimation error***

$$E\left(\begin{bmatrix} X_0 - \hat{X}_0 \\ F_0 - \hat{F}_0 \end{bmatrix} \begin{bmatrix} X_0 - \hat{X}_0 \\ F_0 - \hat{F}_0 \end{bmatrix}^T \right) \leqslant P_0, \tag{5}$$

where $\hat{X}_0$ is the estimate of X_0, $\hat{F}_0 \triangleq \bar{F}(\hat{X}_0, 0)$, and P_0 is a known constant matrix.

A4. ***Variation of uncertainty***

$$E(G_{k,i}^2) \leqslant \overline{q}_{k,i}, i = 1, 2, \cdots, l, \forall k \geqslant 0,$$

where $\{\overline{q}_{k,i}\}_{k=0}^\infty$ is known and uniformly bounded [1)].

扩张状态卡尔曼滤波：算法设计与参数调节

$$\begin{bmatrix} \hat{X}_{k+1} \\ \hat{F}_{k+1} \end{bmatrix} = A_k \begin{bmatrix} \hat{X}_k \\ \hat{F}_k \end{bmatrix} + B_k \hat{G}_k - \boxed{K_k} \left(Y_k - C_k \begin{bmatrix} \hat{X}_k \\ \hat{F}_k \end{bmatrix} \right), \tag{25}$$

$$K_k = -A_k P_k C_k^T \left(C_k P_k C_k^T + \frac{1}{1+\theta} R_k \right)^{-1}, \tag{26}$$

$$P_{k+1} = (1+\theta)(A_k + K_k C_k) P_k (A_k + K_k C_k)^T + K_k R_k K_k{}^T + \left(1 + \frac{1}{\theta}\right) Q_{1,k} + Q_{2,k}, \tag{27}$$

$$Q_{1,k} = \begin{bmatrix} 0_{n\times n} & 0_{n\times l} \\ 0_{l\times n} & 4\overline{Q}_k \end{bmatrix}, Q_{2,k} = \begin{bmatrix} S_k & 0_{n\times l} \\ 0_{l\times n} & 0_{l\times l} \end{bmatrix}, \overline{Q}_k = l \cdot diag\left([\overline{q}_{k,1} \ \overline{q}_{k,2} \ \cdots \ \overline{q}_{k,l}] \right), \tag{28}$$

where

$$\theta = \sqrt{\frac{tr(Q_{1,0})}{tr(P_0)}}, \hat{G}_{k,i} \triangleq sat\left(\bar{G}_{k,i}, \sqrt{\overline{q}_{k,i}} \right), \bar{G}_k \triangleq \bar{G}(\hat{X}_k, k), i = 1, 2, \cdots, l, \tag{29}$$

Compensating the effects caused by uncertain dynamics

W. Xue, W. Bai, etc., ADRC with Adaptive Extended State Observer and Its Application to Air-fuel Ratio Control in Gasoline Engines, ***IEEE Transactions on Industrial Electronics***, 2015.

W. Bai, W. Xue, etc., On extended state based Kalman filter design for a class of nonlinear time-varying uncertain systems, ***SCIENCE CHINA Information Sciences***, 2018.

扩张状态卡尔曼滤波：性能分析

Theorem 2.1 (Stability). If A1-A4 hold, then the estimation error of ESKF, $e_k = \begin{bmatrix} X_k \\ F_k \end{bmatrix} - \begin{bmatrix} \hat{X}_k \\ \hat{F}_k \end{bmatrix}$, satisfies

$$E(e_k e_k^T) \leqslant P_k, \forall k \geqslant 0, \tag{30}$$

and $\{P_k\}_{k=0}^{\infty}$ is uniformly bounded. Then, there exists a positive matrix P^* such that

$$P_k \leqslant P^*, \forall k \geqslant 0. \tag{31}$$

➢**Stability: The estimation error is bounded in mean square.**

➢**Consistence:** $E(e_k e_k^T) \leq P_k$ **and** P_k **is timely available.**

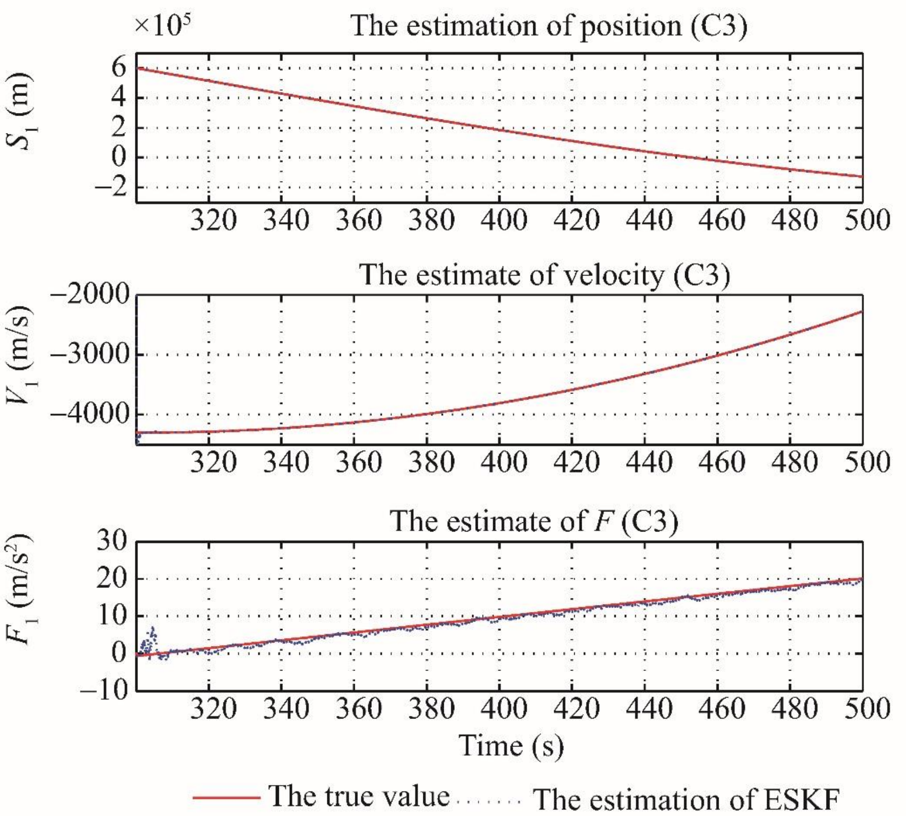
Theorem 2.2 (Optimality). If $F(X_k, k) = F_0$, $\bar{A}_k = \bar{A}$, $\bar{B}_k = \bar{B}, \bar{C}_k = \bar{C}, R_k = R, S_k = S, \forall k \geqslant 0$, then ESKF (25)-(28) with $Q_k = 0$ is an asymptotically unbiased minimum variance estimation for $\begin{bmatrix} X_k^T & F_k^T \end{bmatrix}^T$ among all the functions of $\{Y_i\}_{i=1}^{k-1}$.

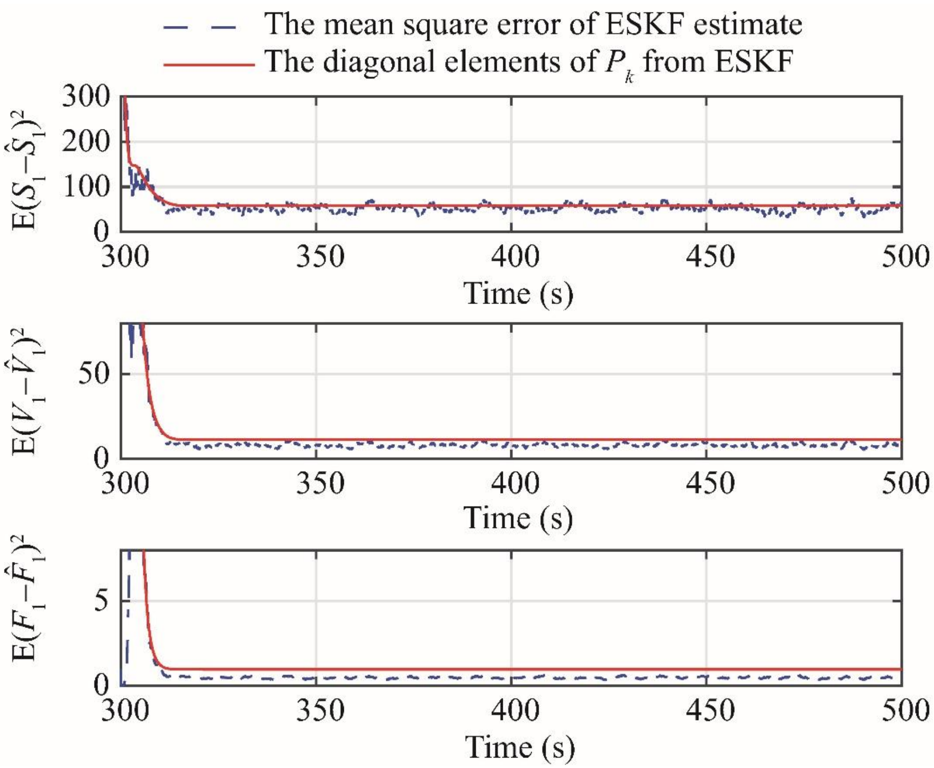
IEEE Transactions on Industrial Electronics, 2015; ***SCIENCE CHINA Information Sciences***, 2018.

具有未知加速度的空间目标动力学

$$\begin{cases} \dot{S}(t) = V(t), \\ \dot{V}(t) = -2\Omega V(t) - \Omega^2 S(t) - \dfrac{\mu}{\|S(t)\|^3} S(t) + \Delta_F(t), \\ Y_k = \bar{C}_k S_k + v_k, \end{cases}$$

未知非线性动态





谢谢！